

BFS Traversal

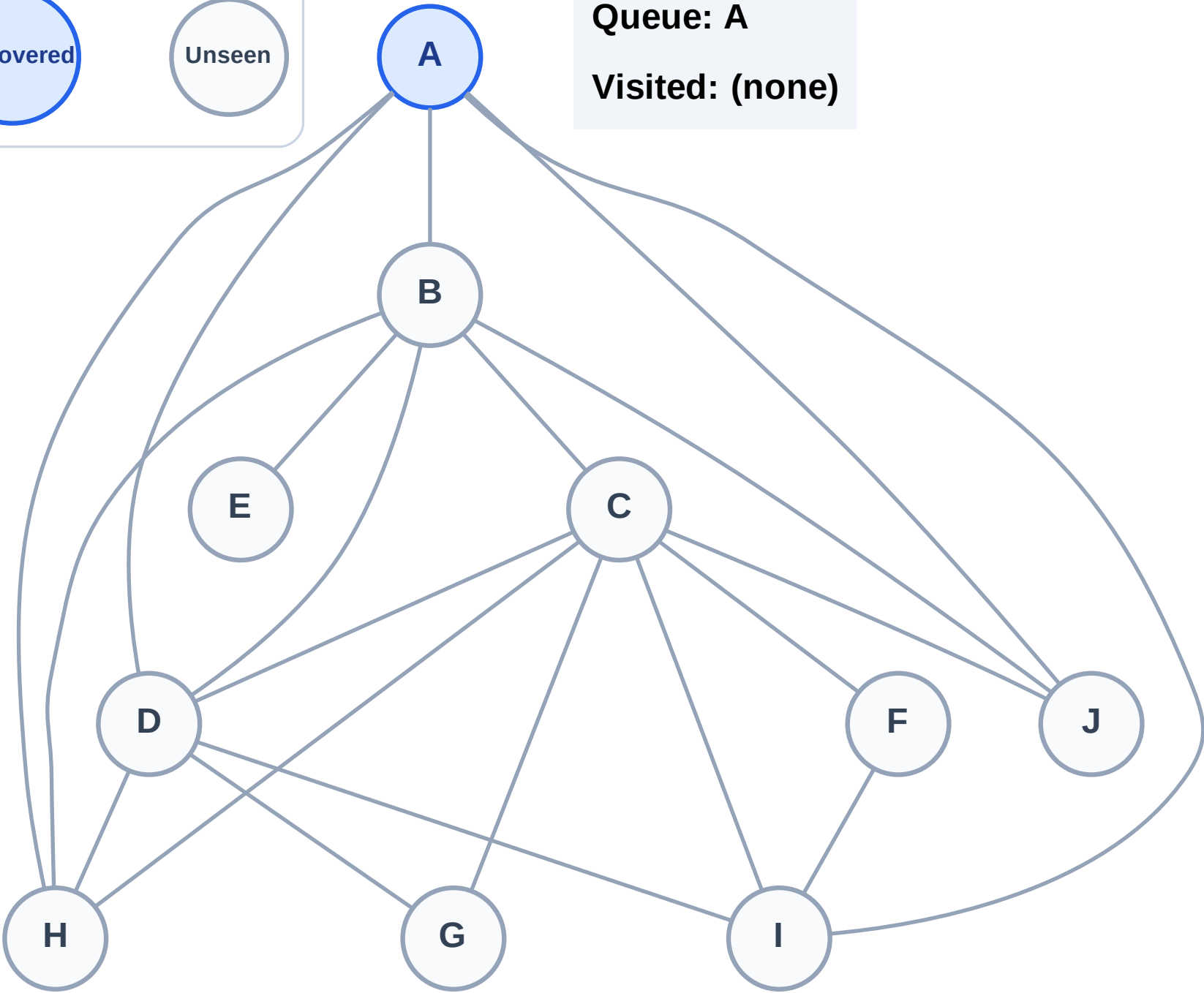
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

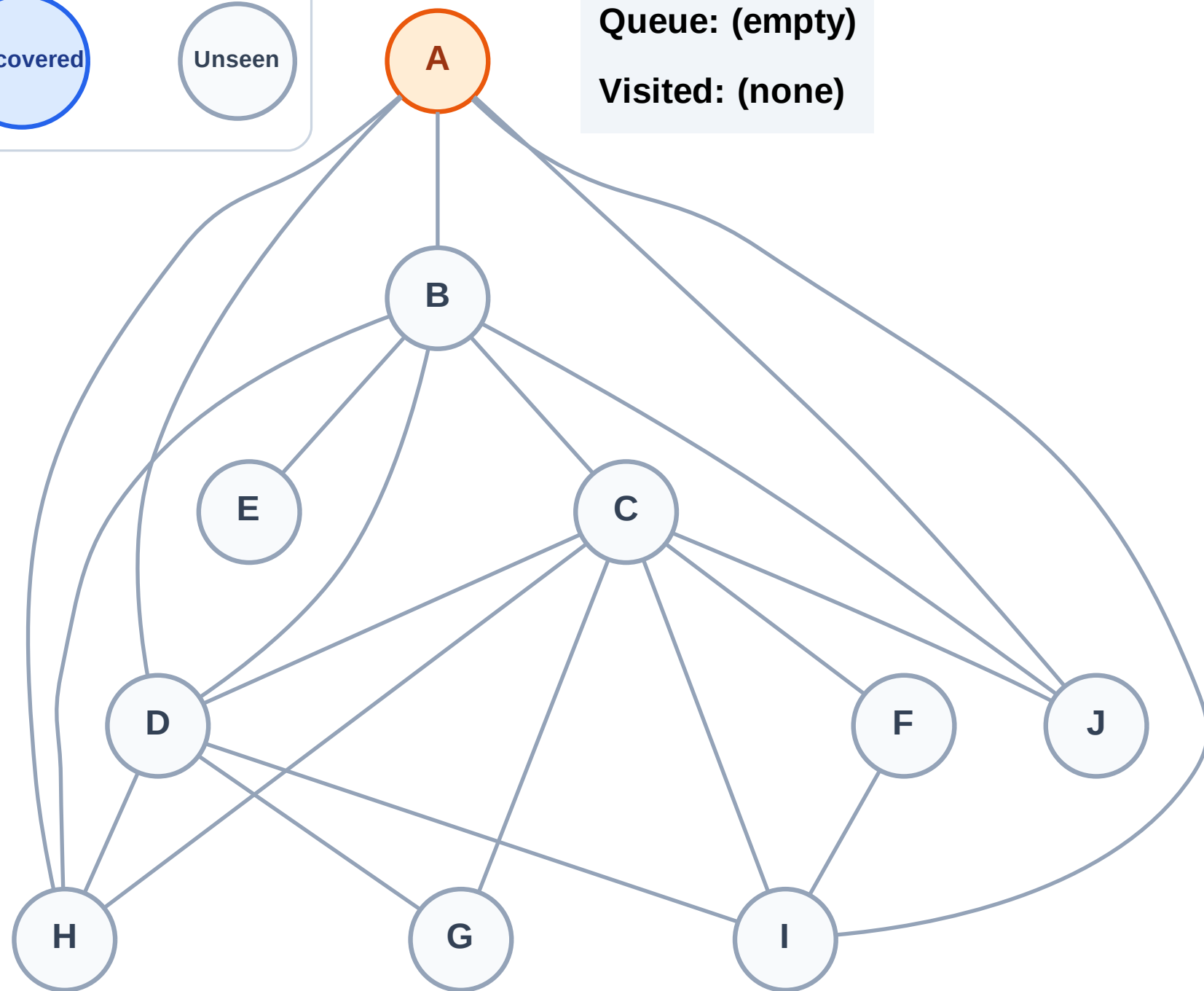
Step 2: Process node A

Legend



Queue: (empty)

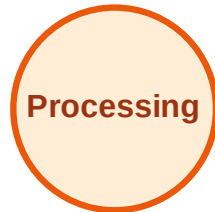
Visited: (none)



BFS Traversal

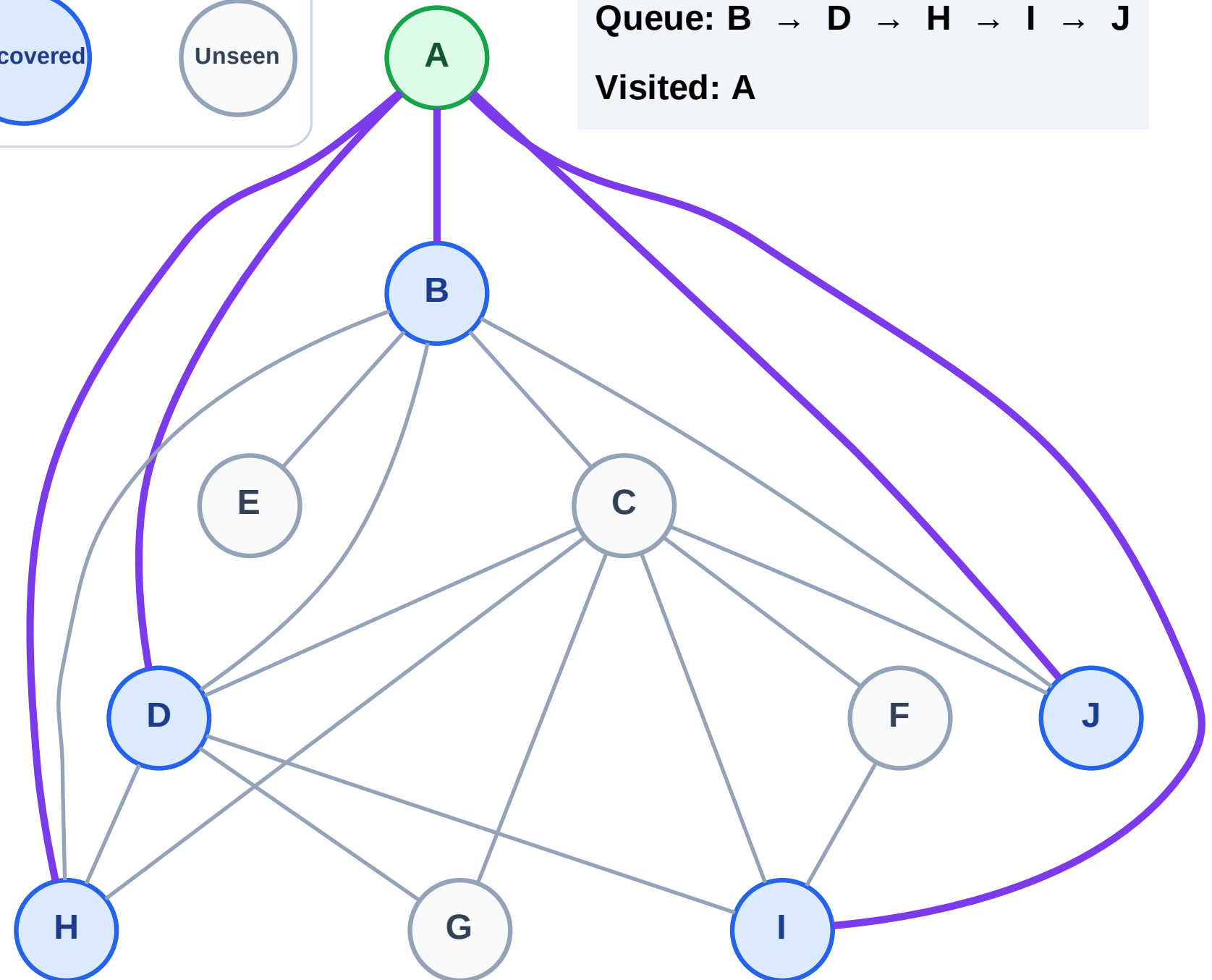
Step 3: Discover B, D, H, I, J from A

Legend



Queue: B → D → H → I → J

Visited: A



BFS Traversal

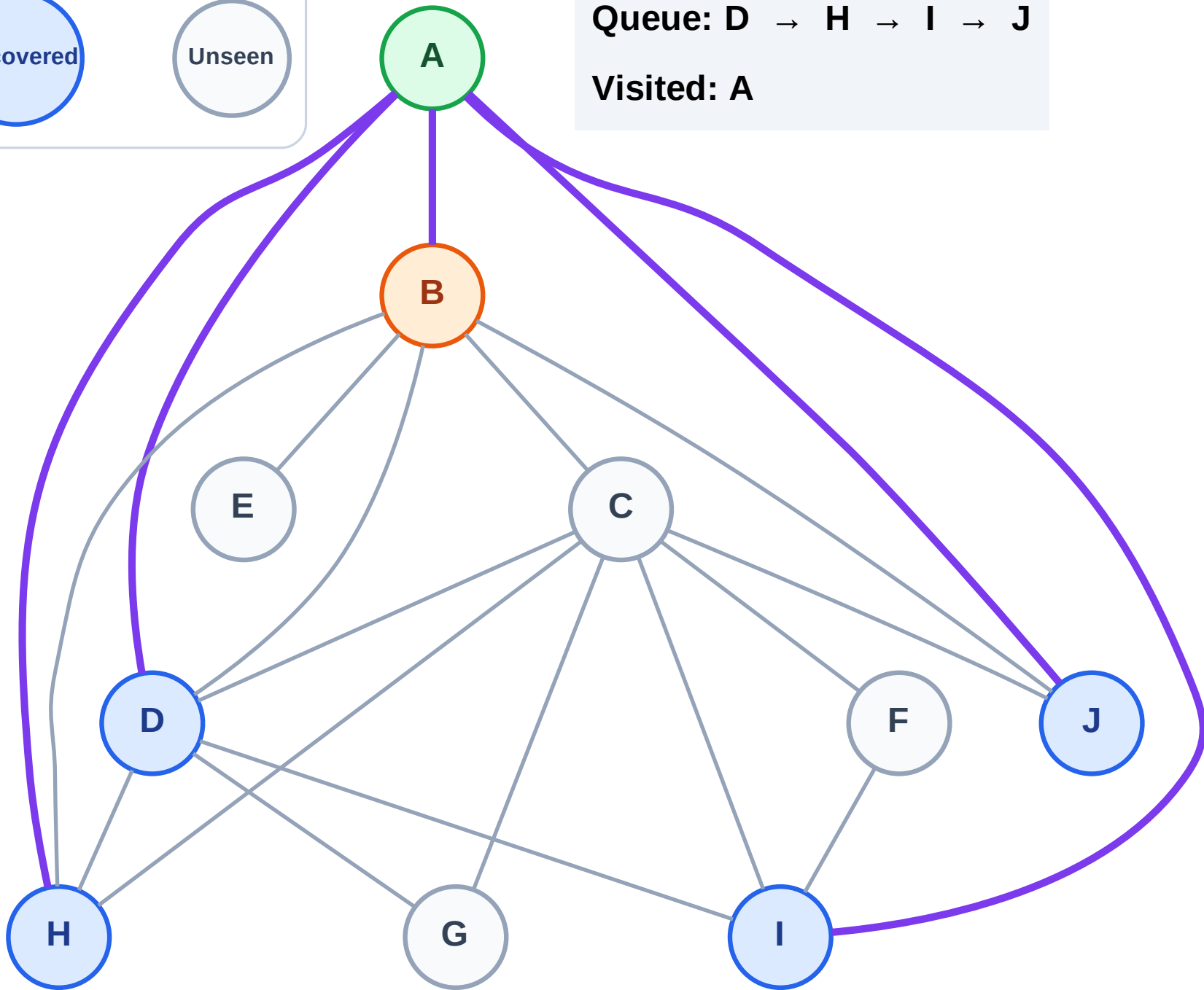
Step 4: Process node B

Legend



Queue: D → H → I → J

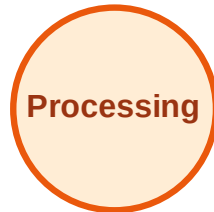
Visited: A



BFS Traversal

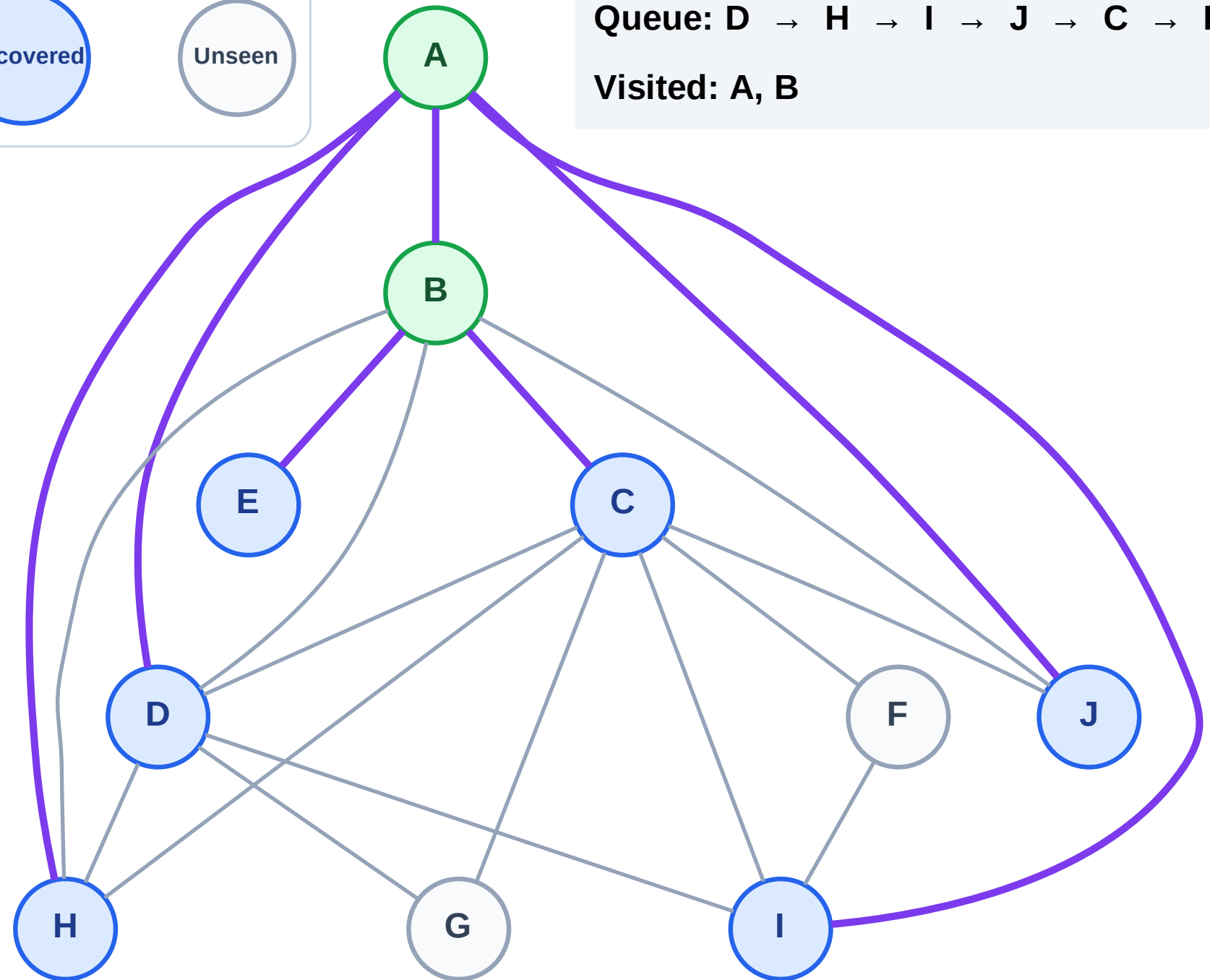
Step 5: Discover C, E from B

Legend



Queue: D → H → I → J → C → E

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

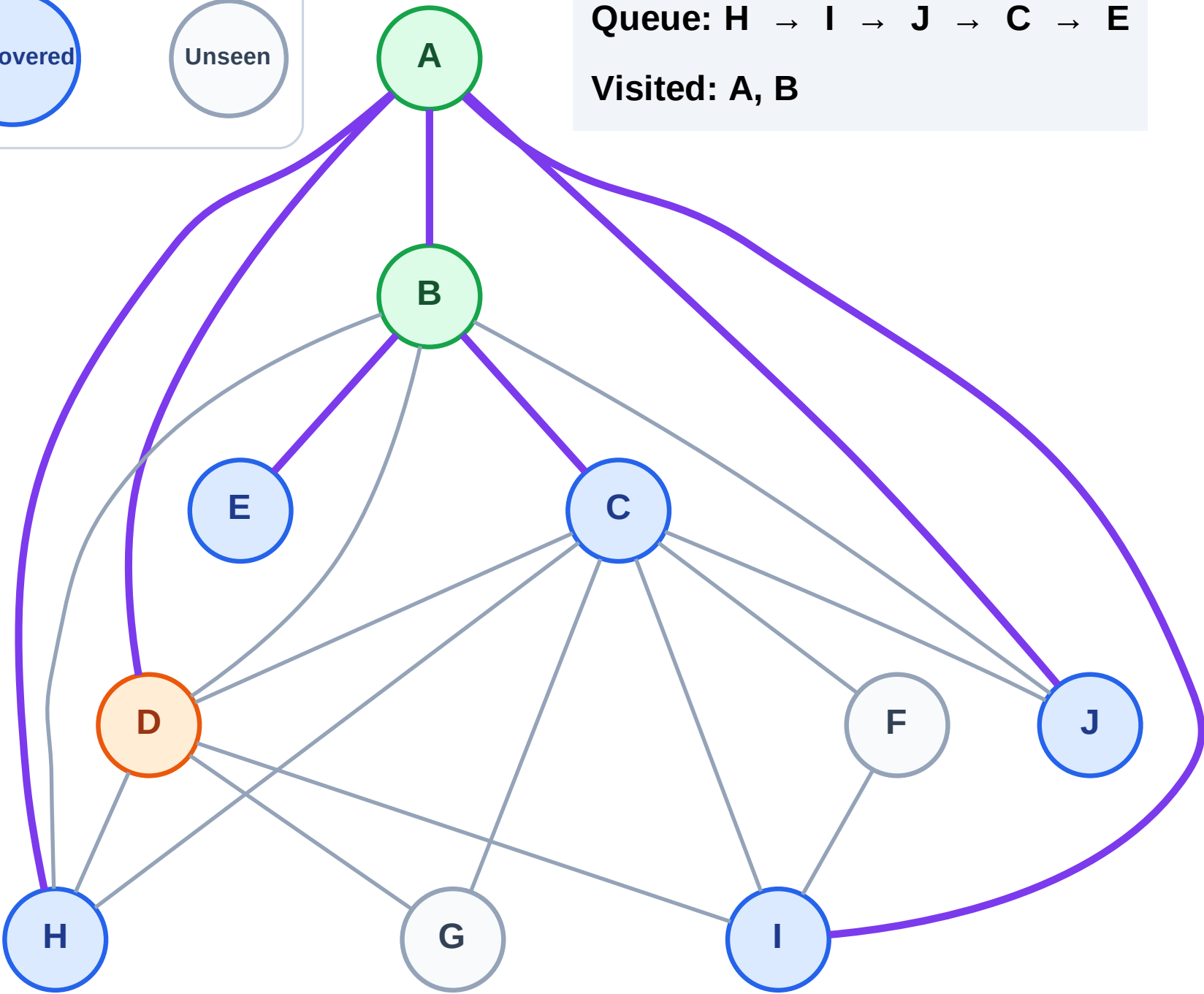
Processing

Discovered

Unseen

Queue: H → I → J → C → E

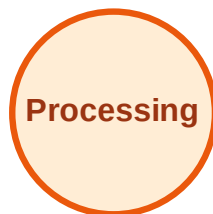
Visited: A, B



BFS Traversal

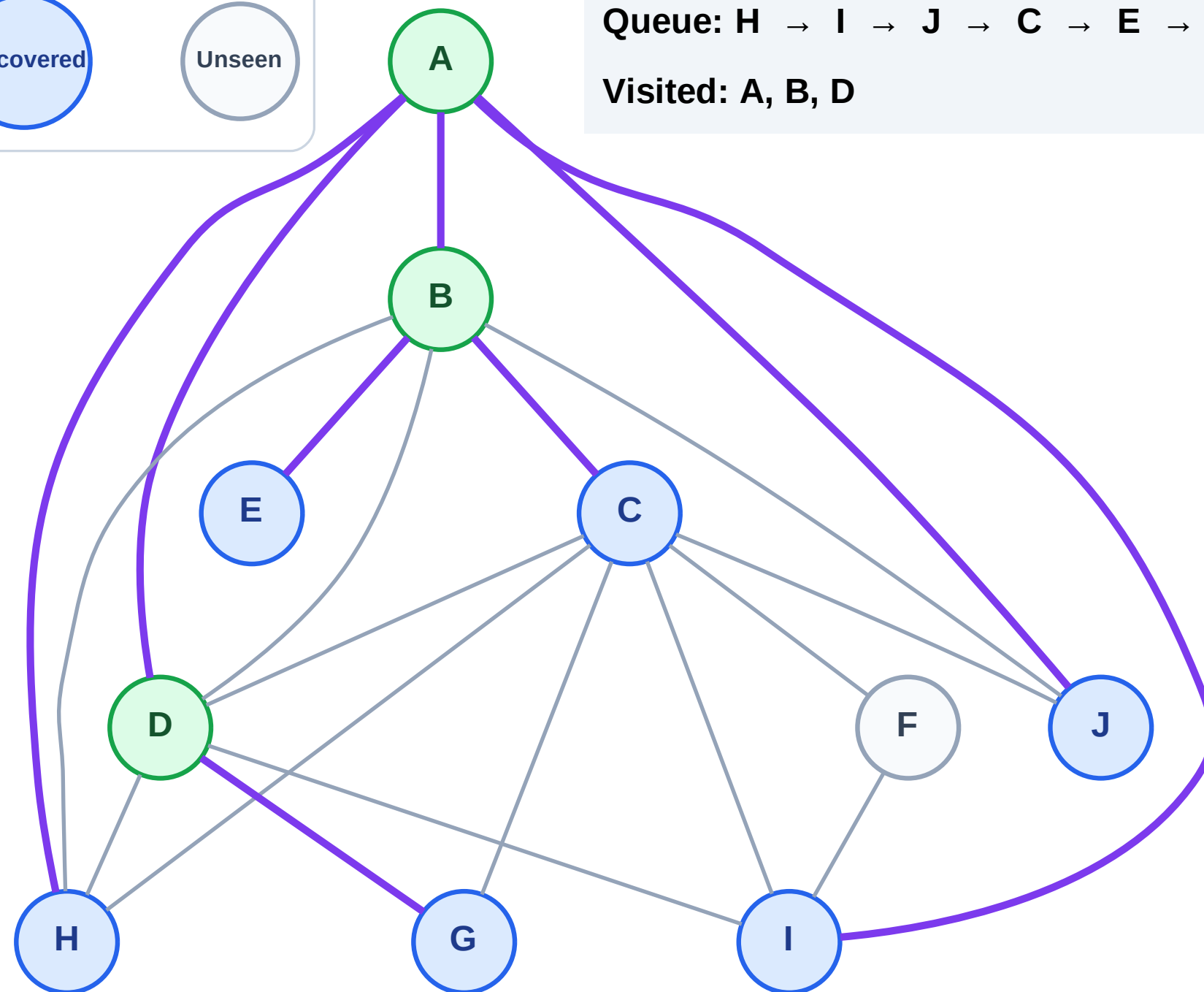
Step 7: Discover G from D

Legend



Queue: H → I → J → C → E → G

Visited: A, B, D



BFS Traversal

Step 8: Process node H

Legend

Complete

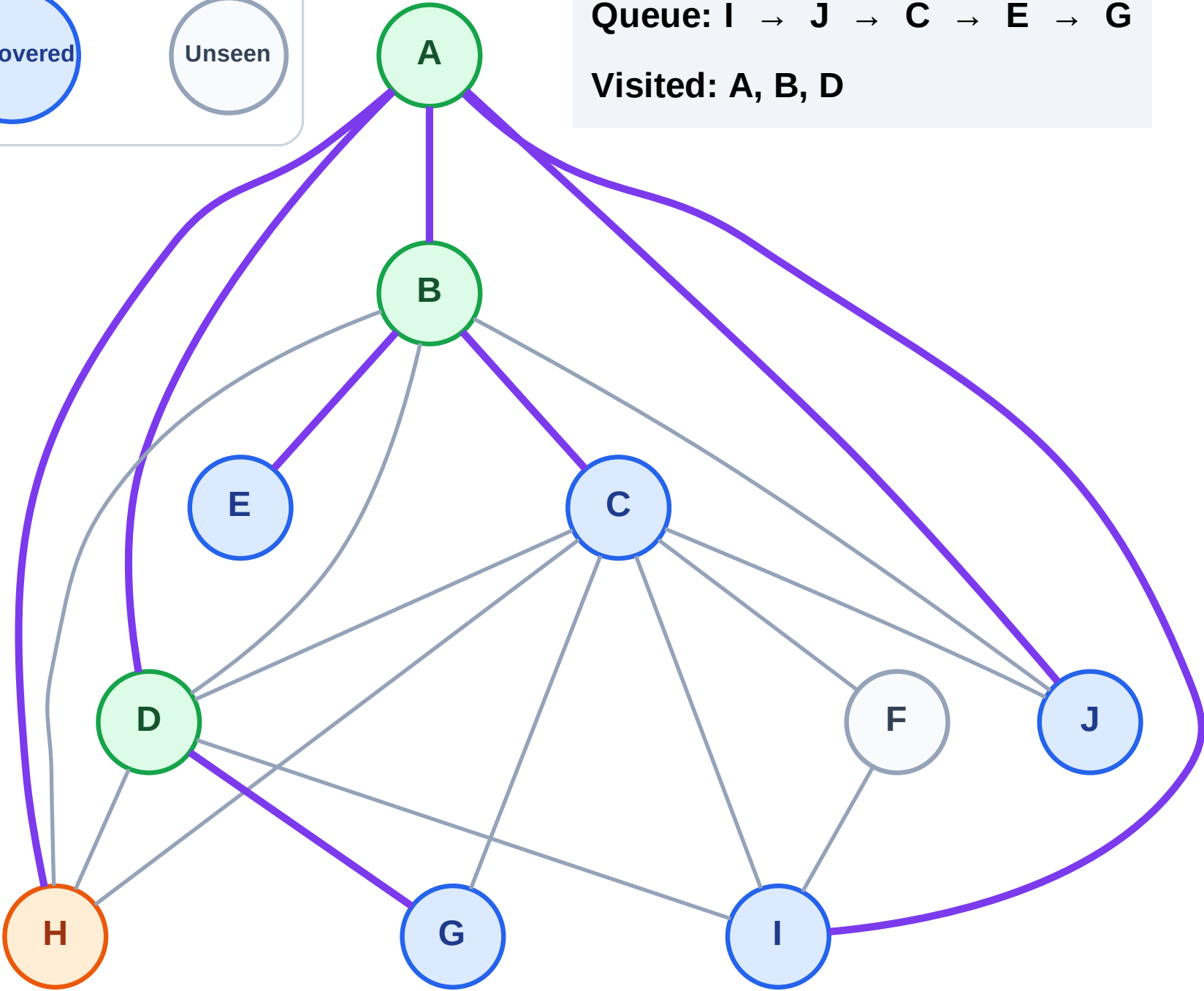
Processing

Discovered

Unseen

Queue: I → J → C → E → G

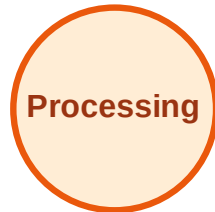
Visited: A, B, D



BFS Traversal

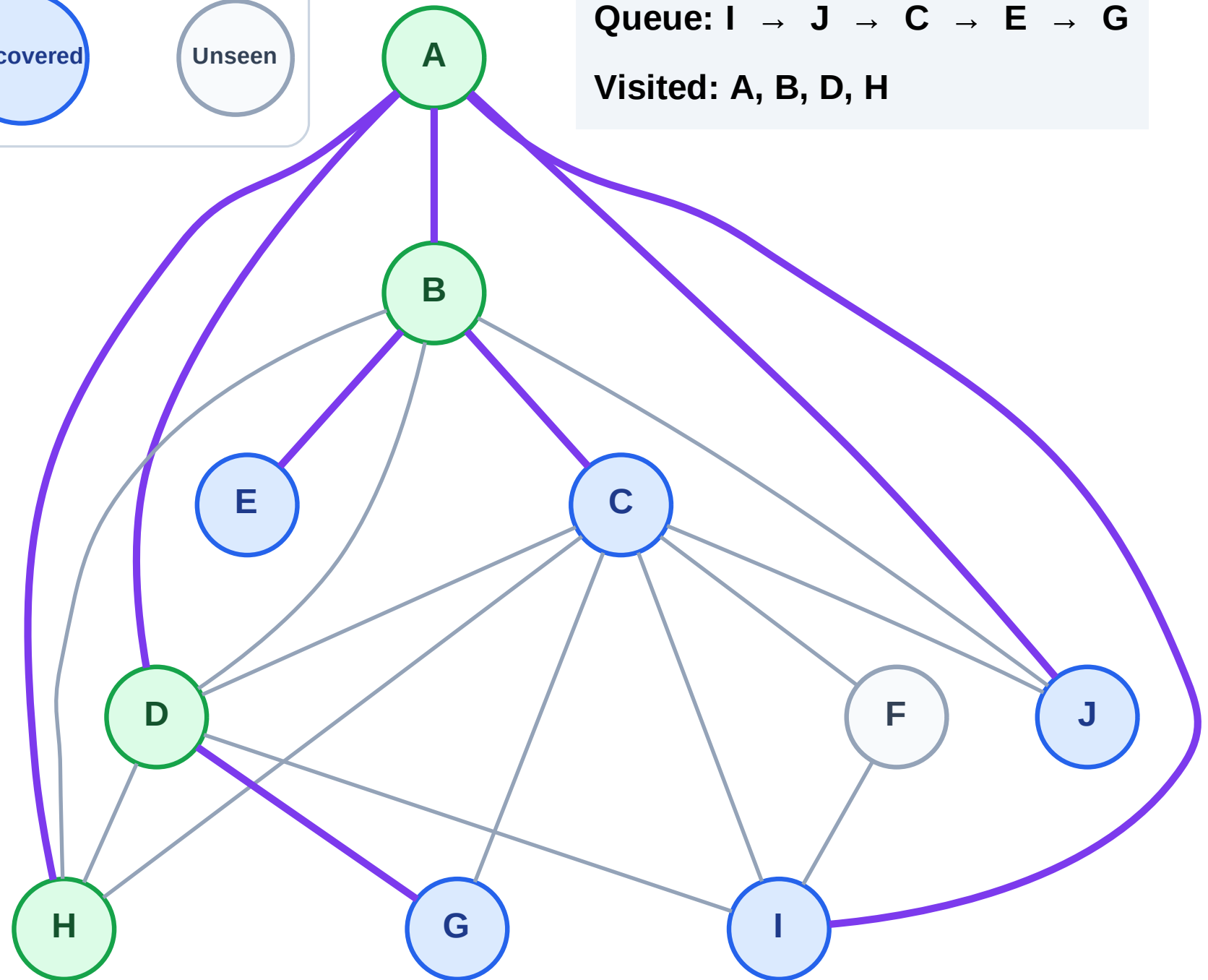
Step 9: No new neighbors from H

Legend



Queue: I → J → C → E → G

Visited: A, B, D, H



BFS Traversal

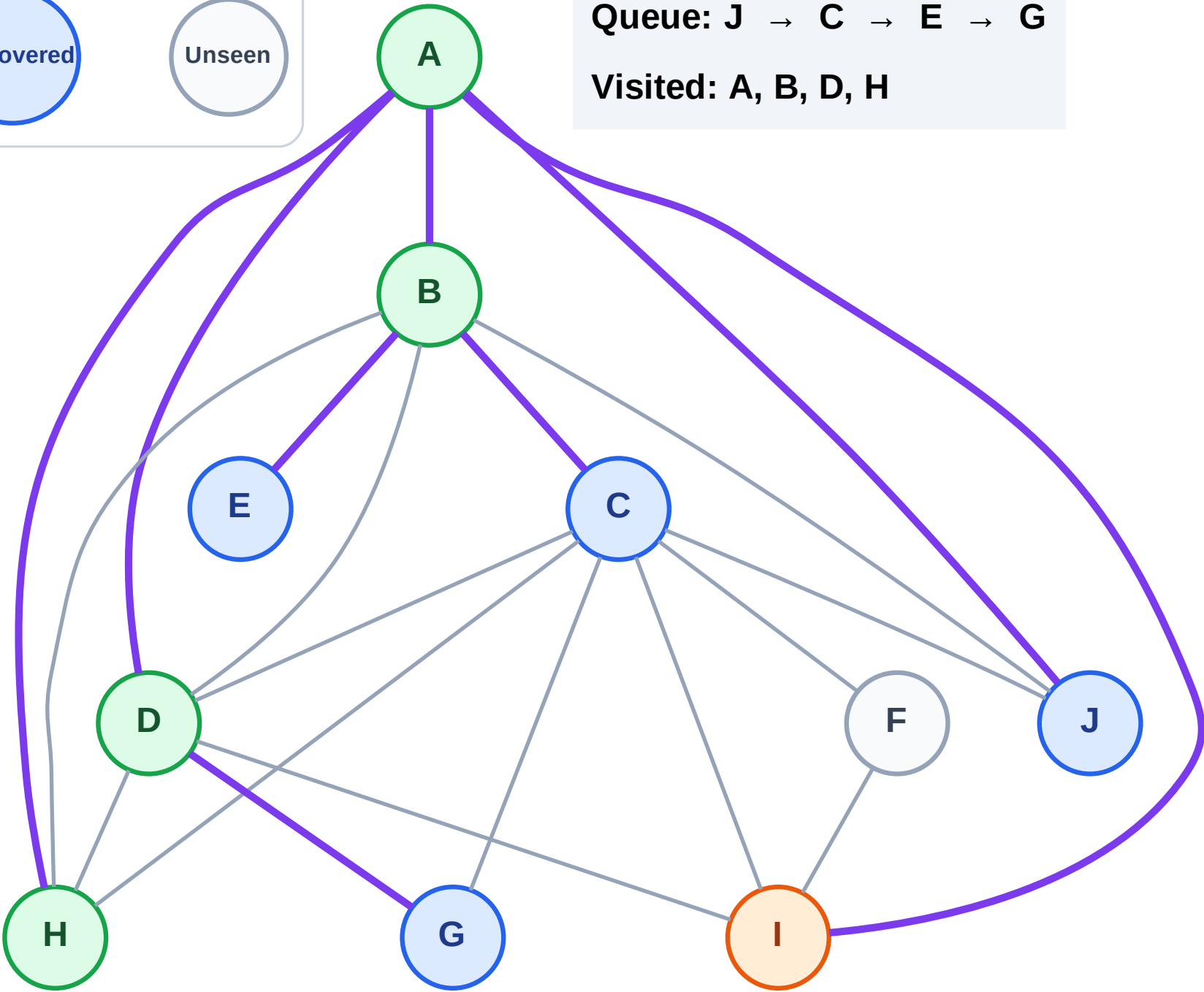
Step 10: Process node I

Legend



Queue: J → C → E → G

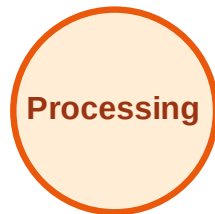
Visited: A, B, D, H



BFS Traversal

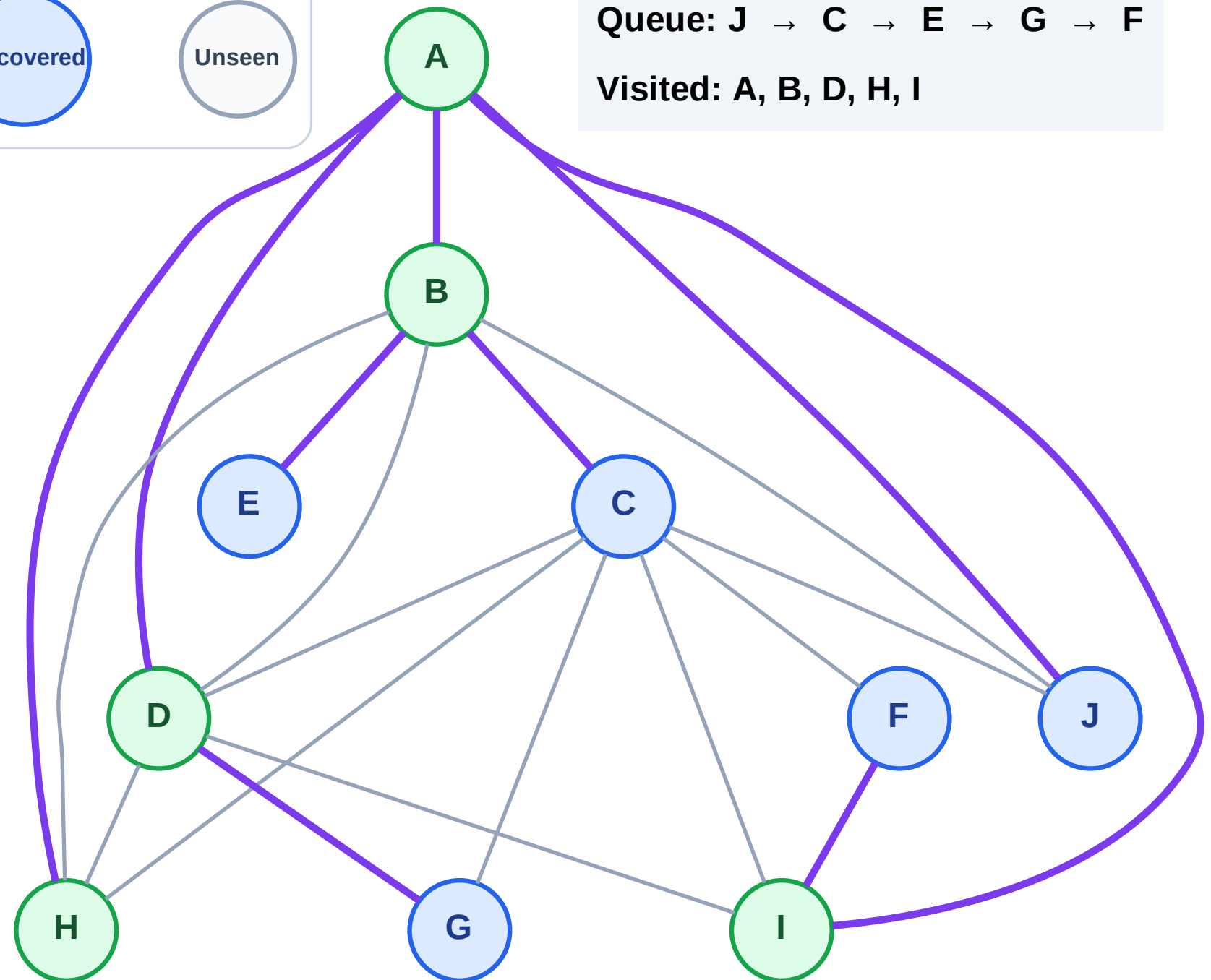
Step 11: Discover F from I

Legend



Queue: J → C → E → G → F

Visited: A, B, D, H, I



BFS Traversal

Step 12: Process node J

Legend

Complete

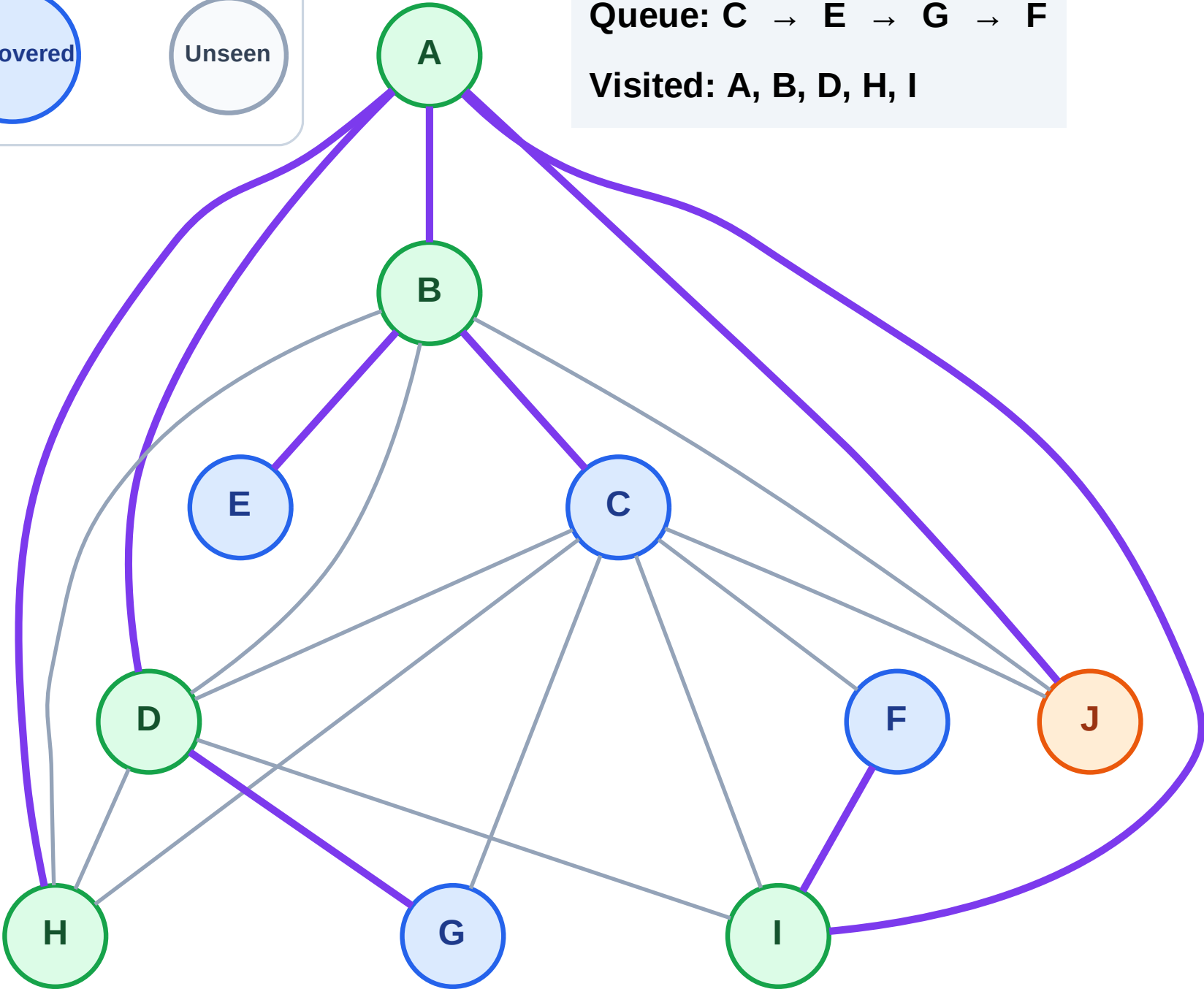
Processing

Discovered

Unseen

Queue: C → E → G → F

Visited: A, B, D, H, I



BFS Traversal

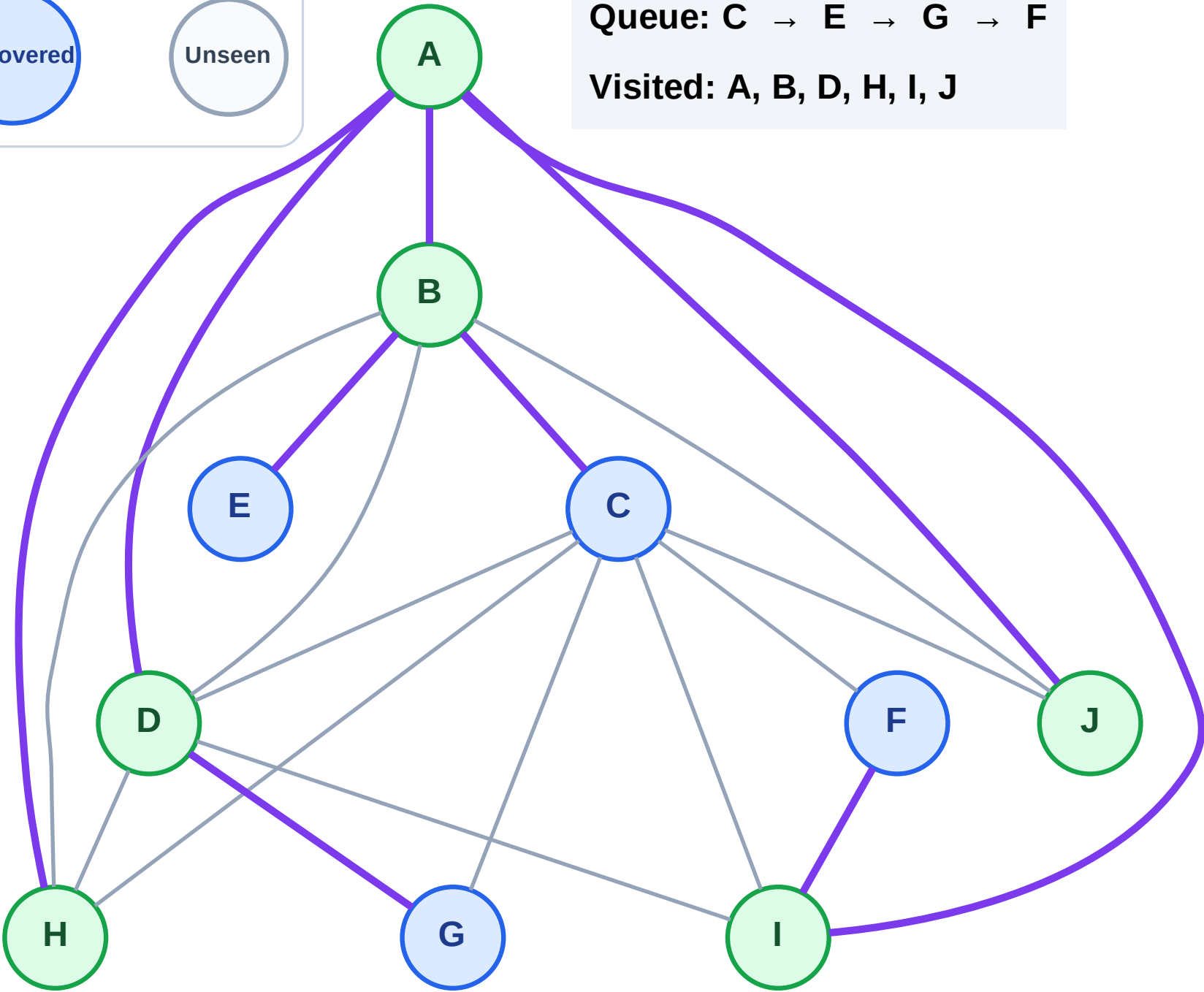
Step 13: No new neighbors from J

Legend



Queue: C → E → G → F

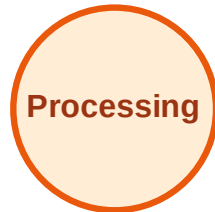
Visited: A, B, D, H, I, J



BFS Traversal

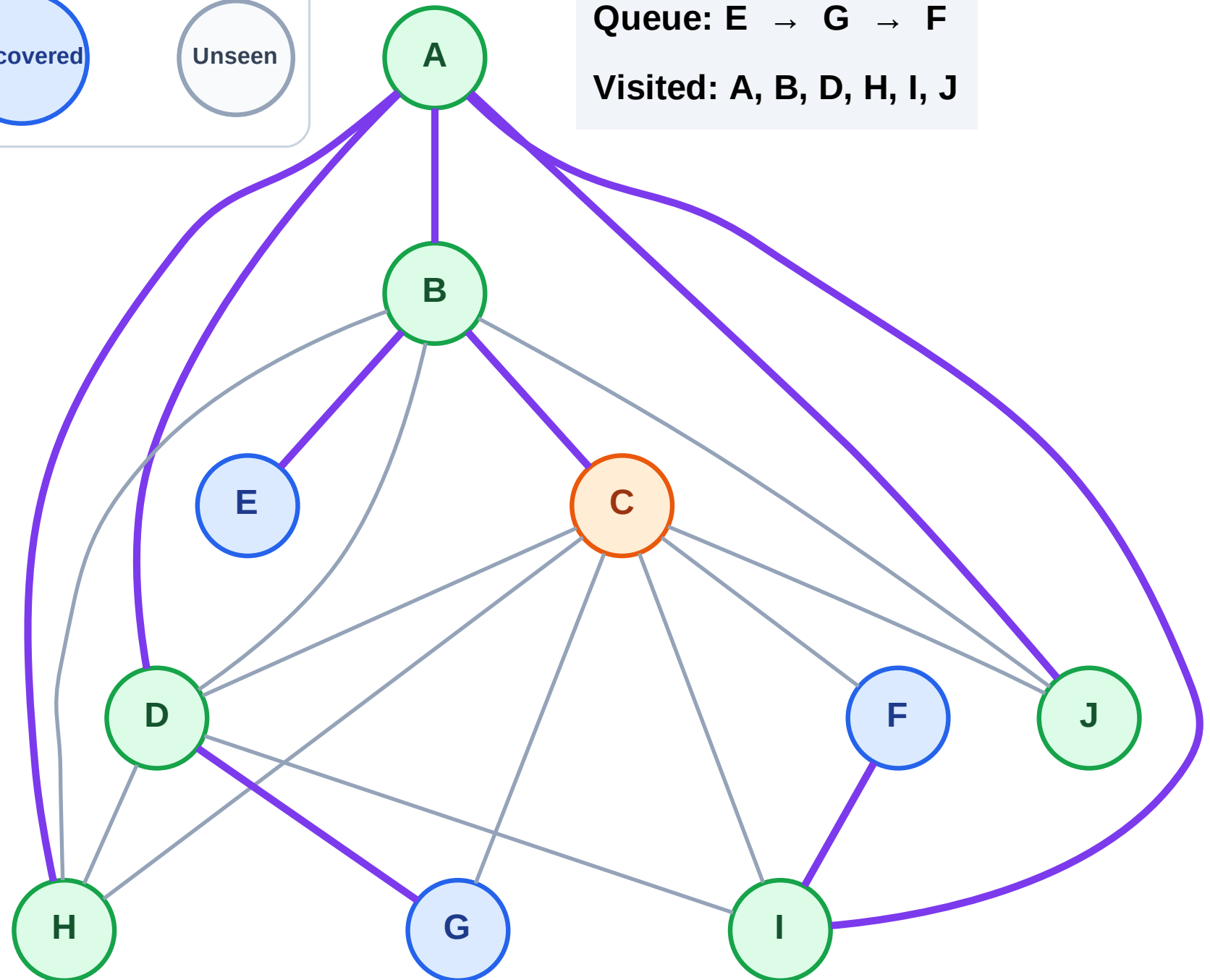
Step 14: Process node C

Legend



Queue: E → G → F

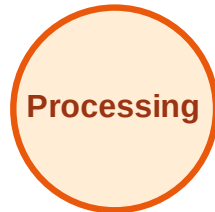
Visited: A, B, D, H, I, J



BFS Traversal

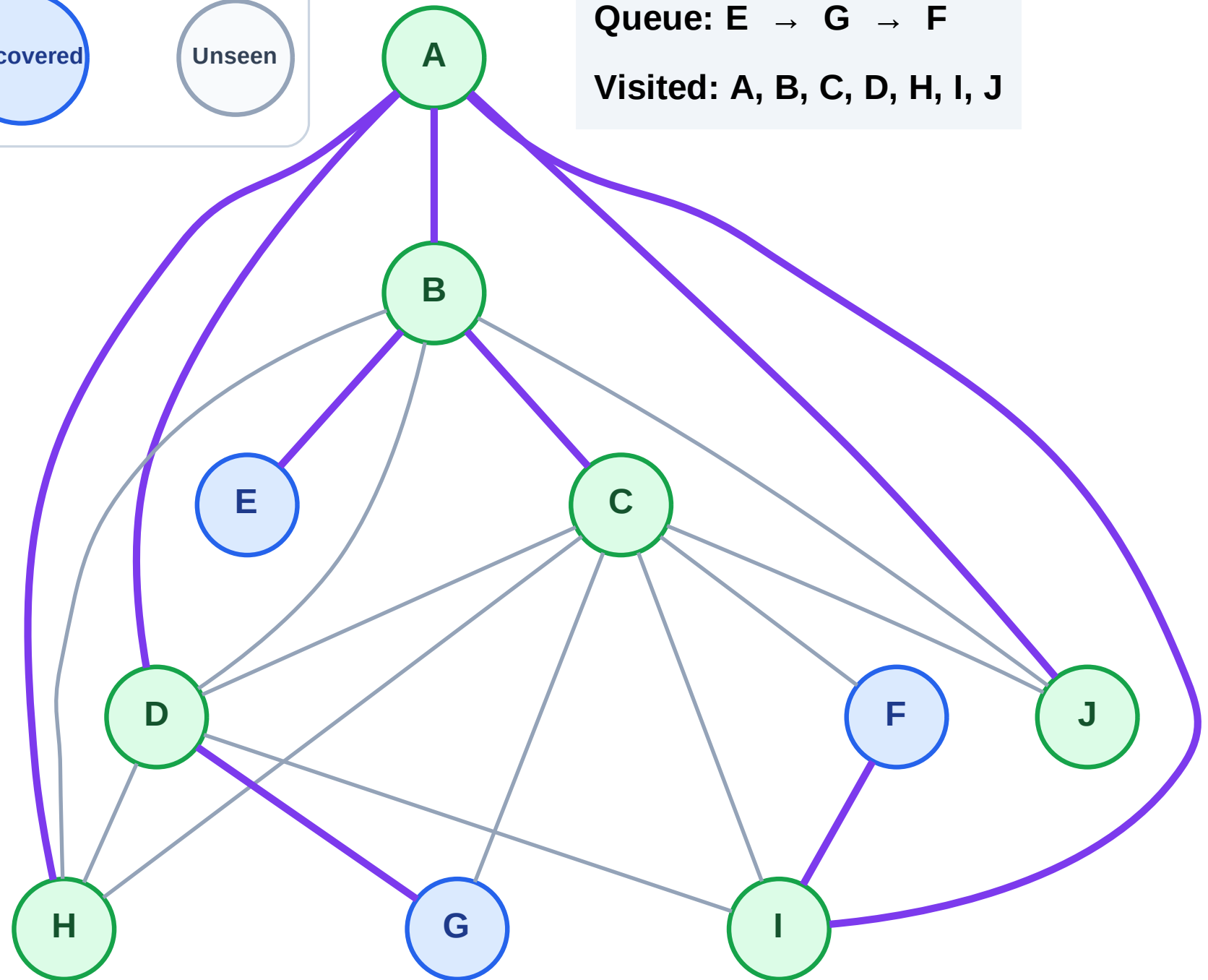
Step 15: No new neighbors from C

Legend



Queue: E → G → F

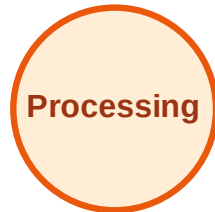
Visited: A, B, C, D, H, I, J



BFS Traversal

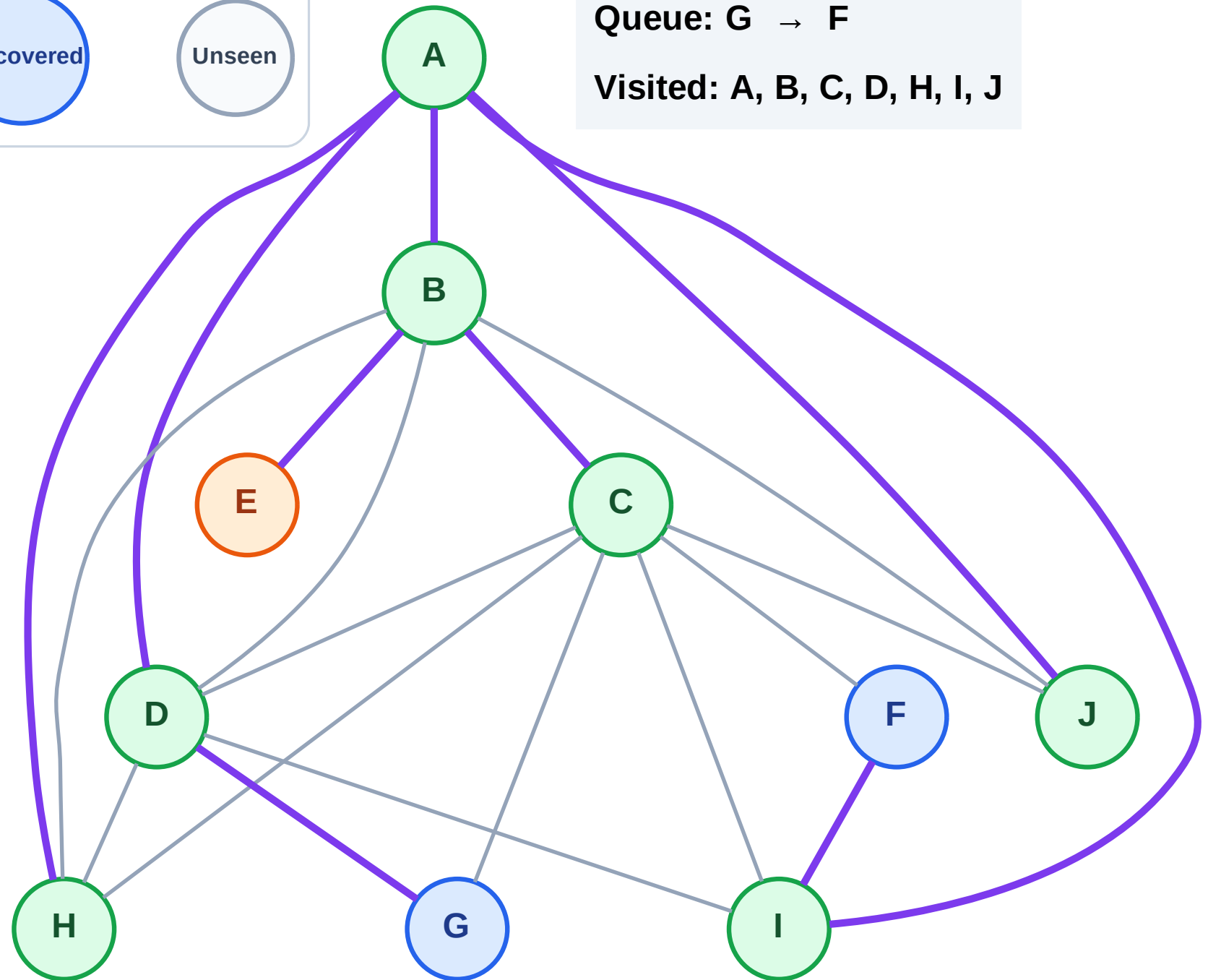
Step 16: Process node E

Legend



Queue: G → F

Visited: A, B, C, D, H, I, J



BFS Traversal

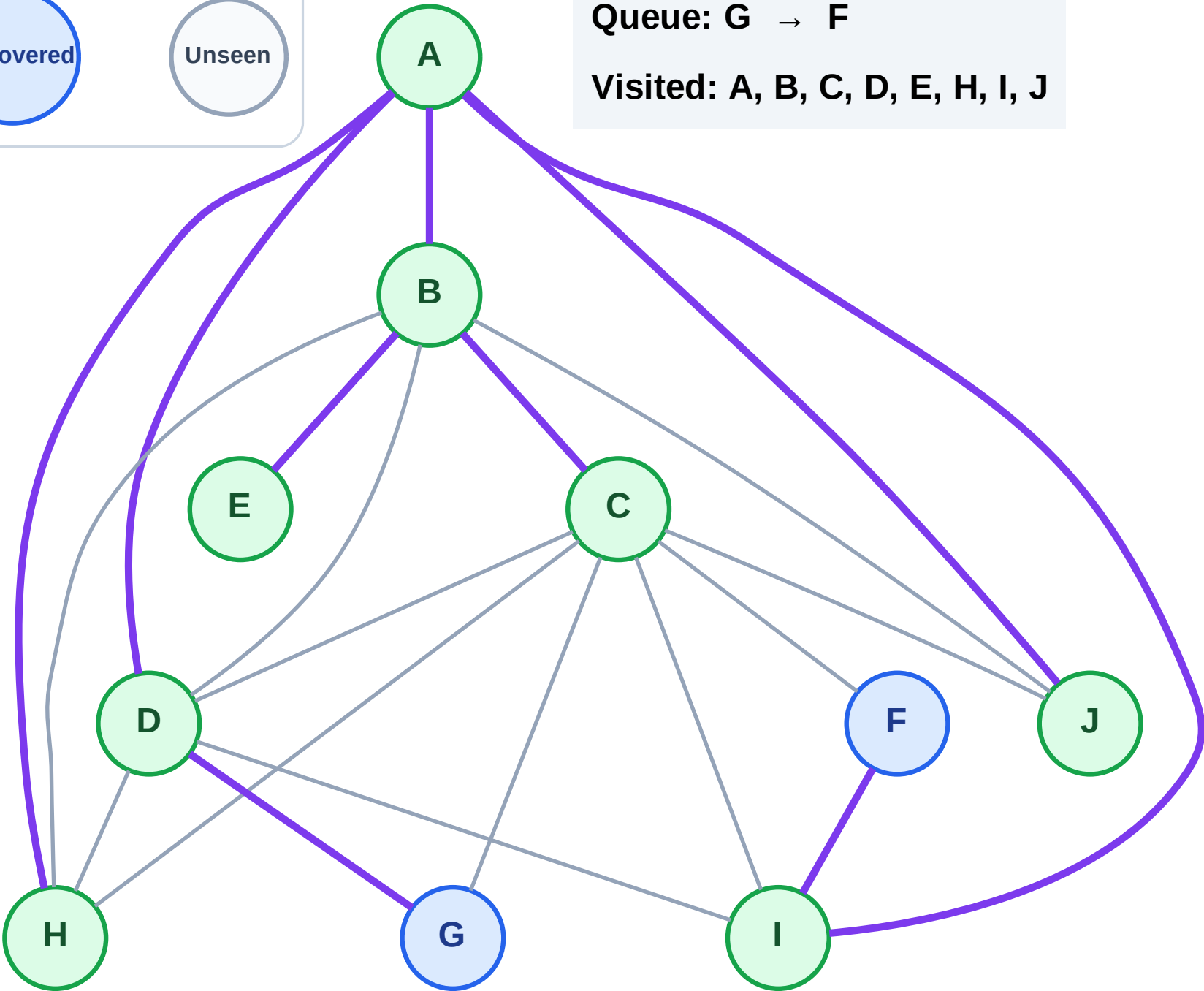
Step 17: No new neighbors from E

Legend



Queue: G → F

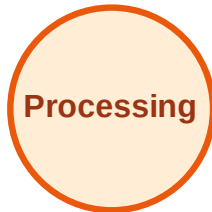
Visited: A, B, C, D, E, H, I, J



BFS Traversal

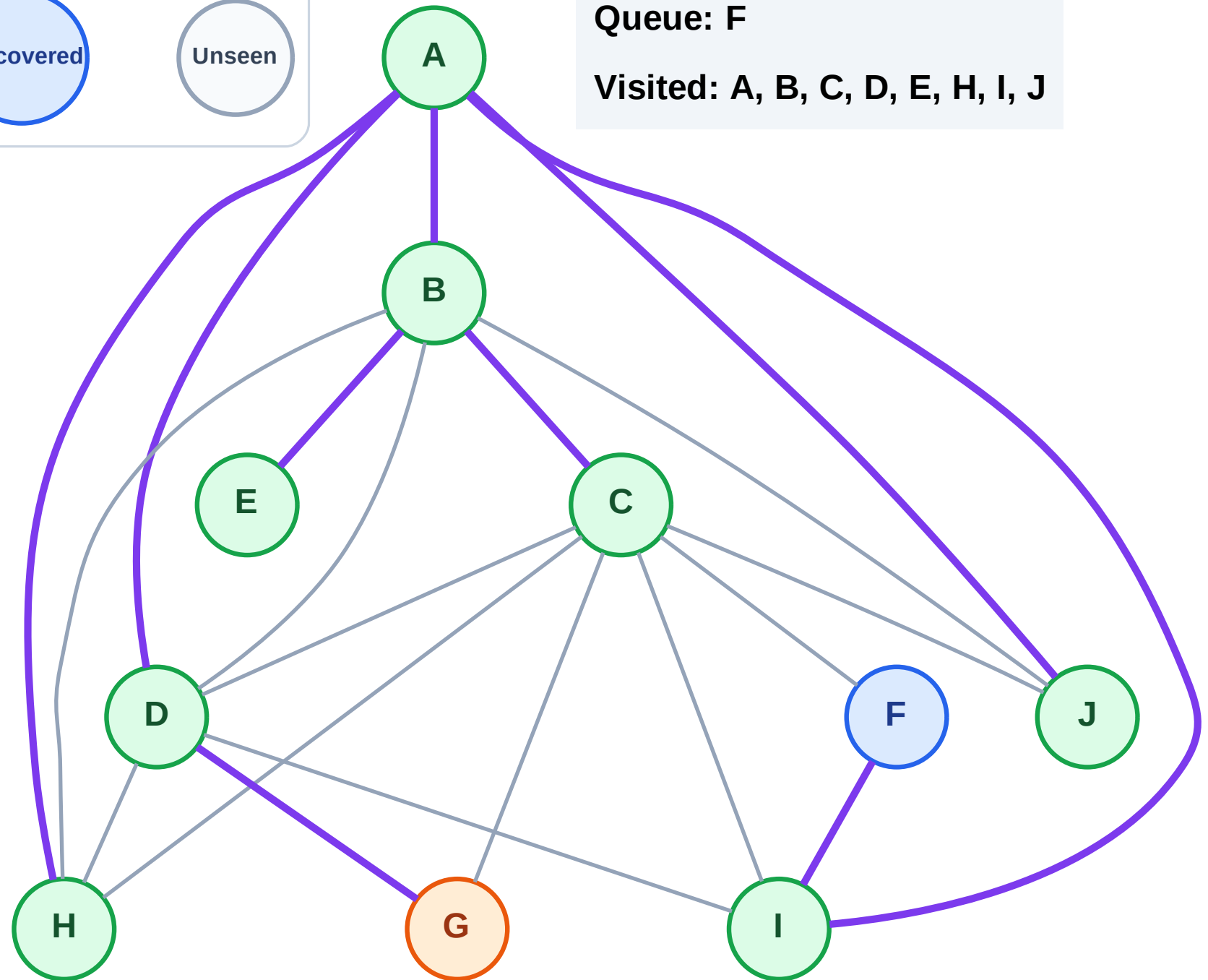
Step 18: Process node G

Legend



Queue: F

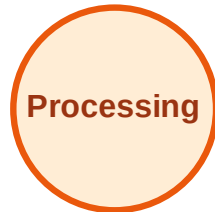
Visited: A, B, C, D, E, H, I, J



BFS Traversal

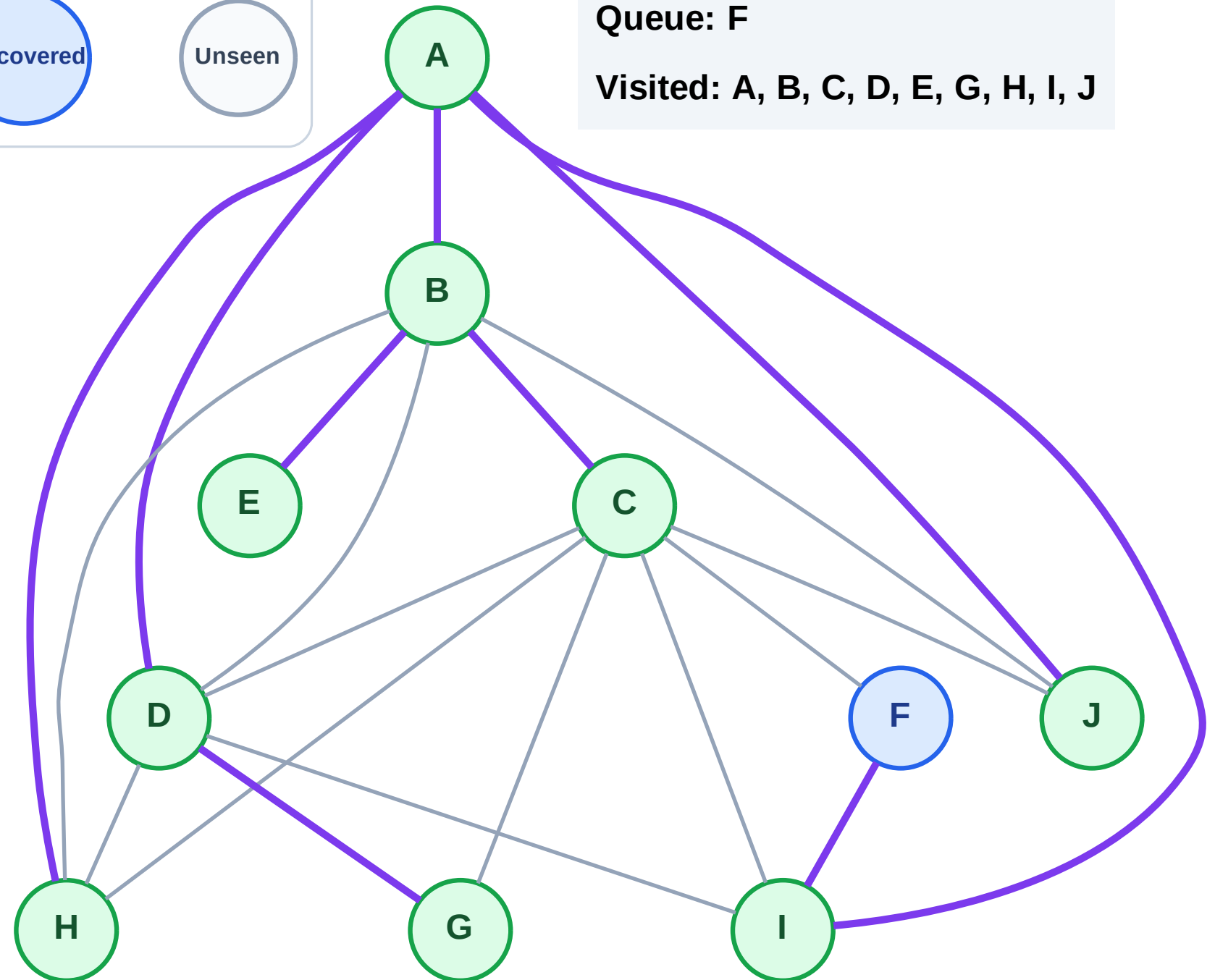
Step 19: No new neighbors from G

Legend



Queue: F

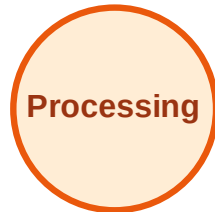
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

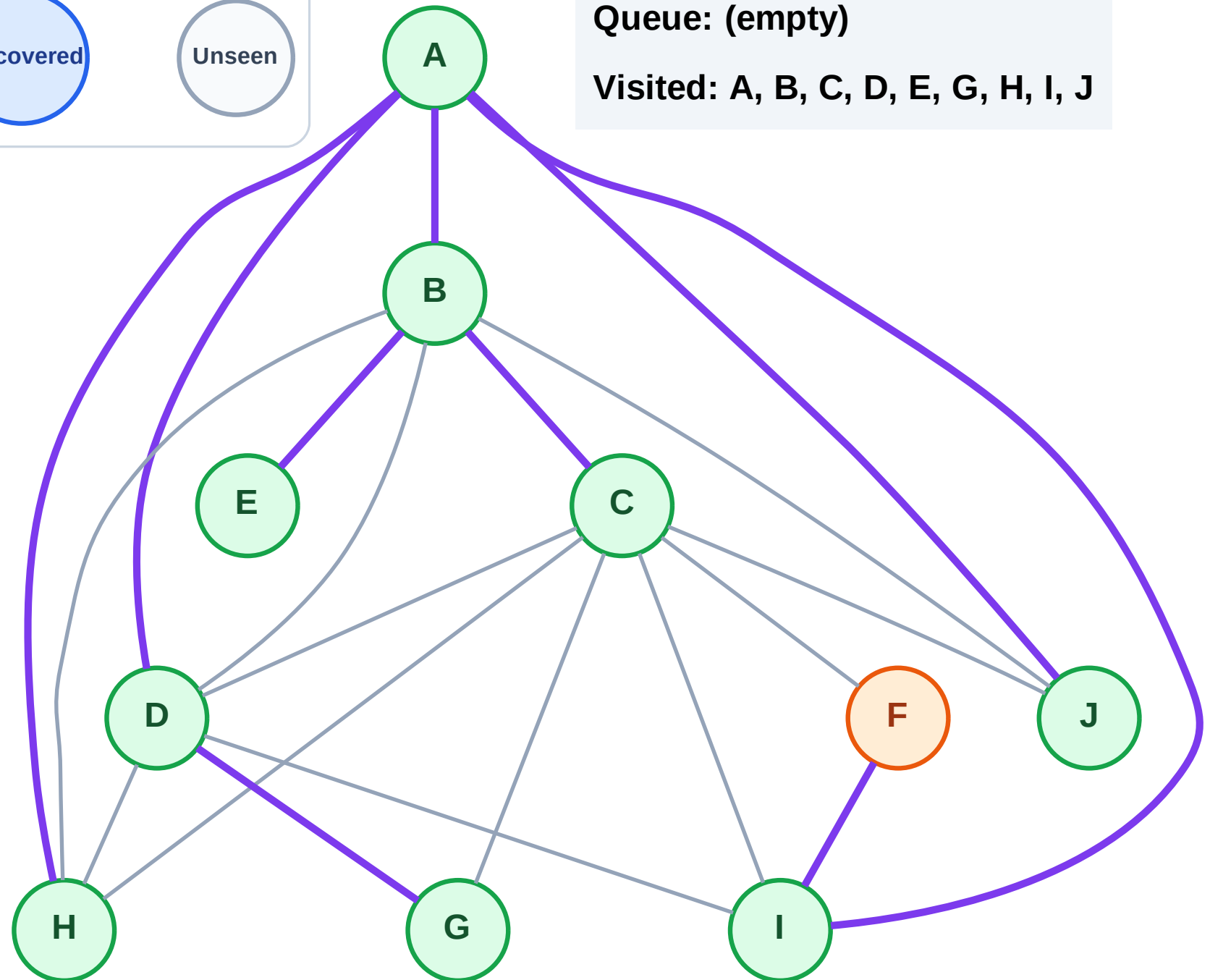
Step 20: Process node F

Legend



Queue: (empty)

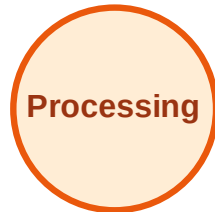
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

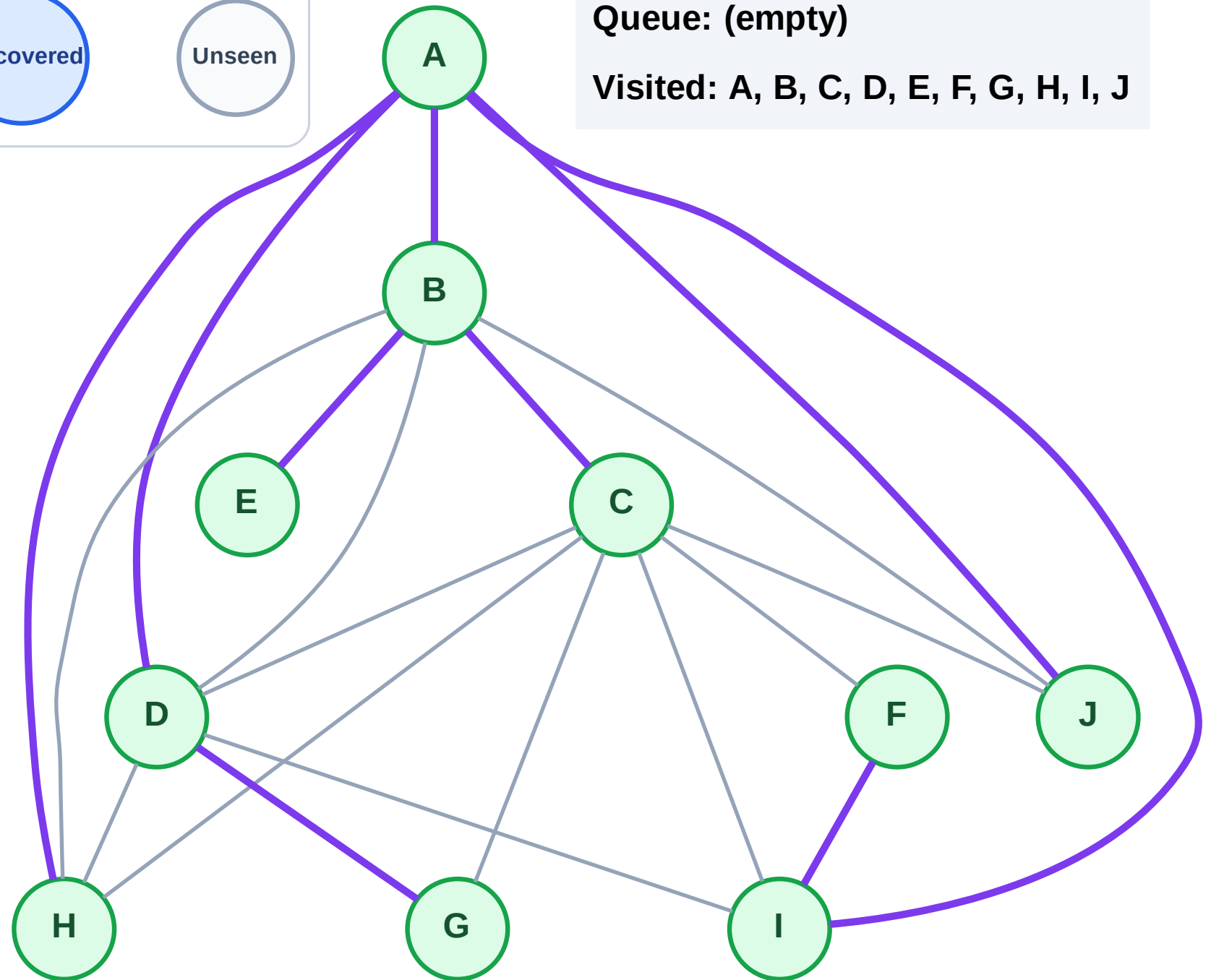
Step 21: No new neighbors from F

Legend



Queue: (empty)

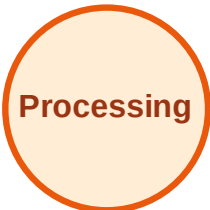
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

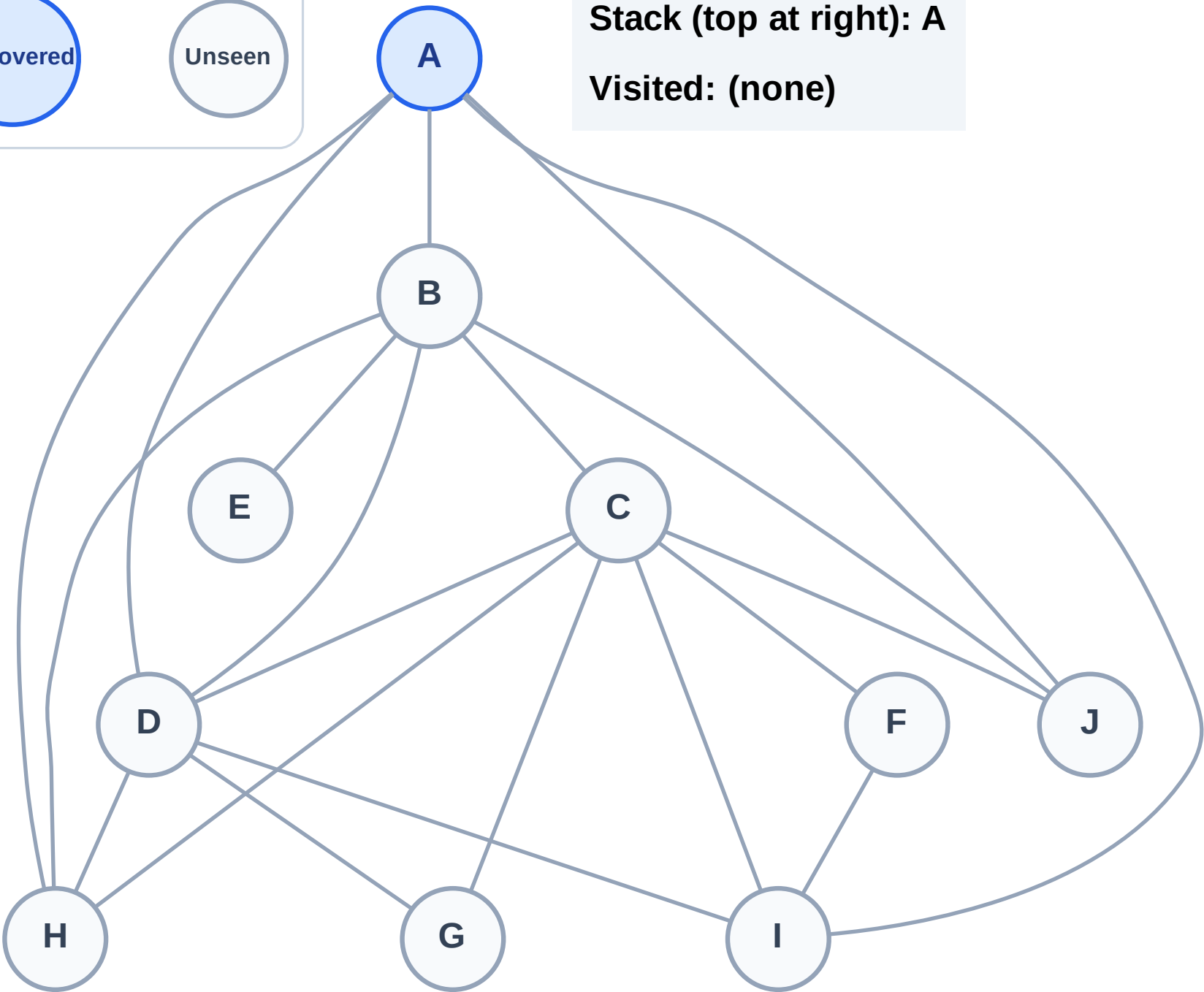
Step 1: Start at A

Legend



Stack (top at right): A

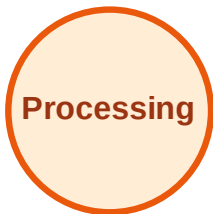
Visited: (none)



DFS Traversal

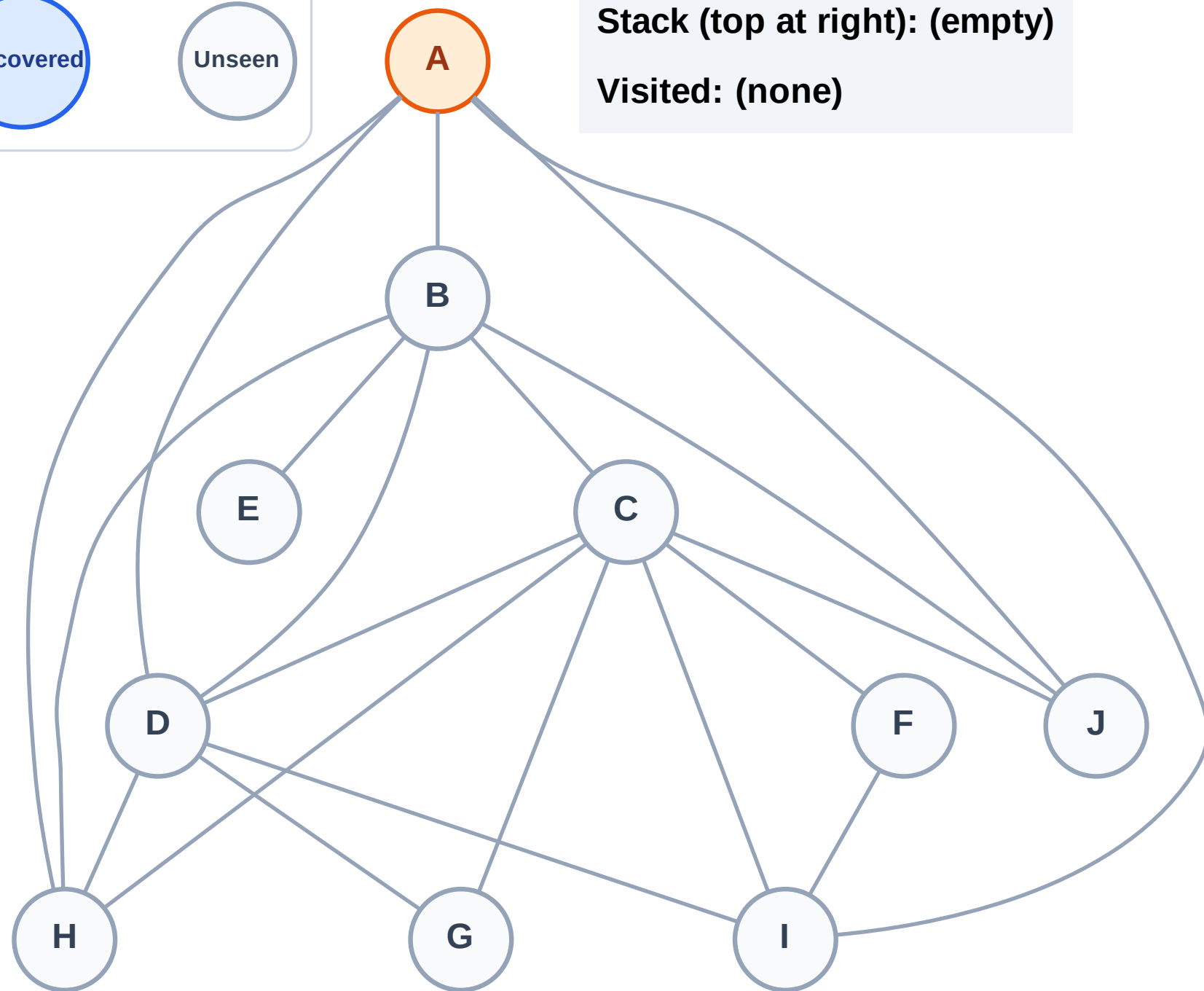
Step 2: Process node A

Legend



Stack (top at right): (empty)

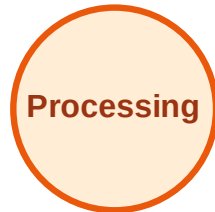
Visited: (none)



DFS Traversal

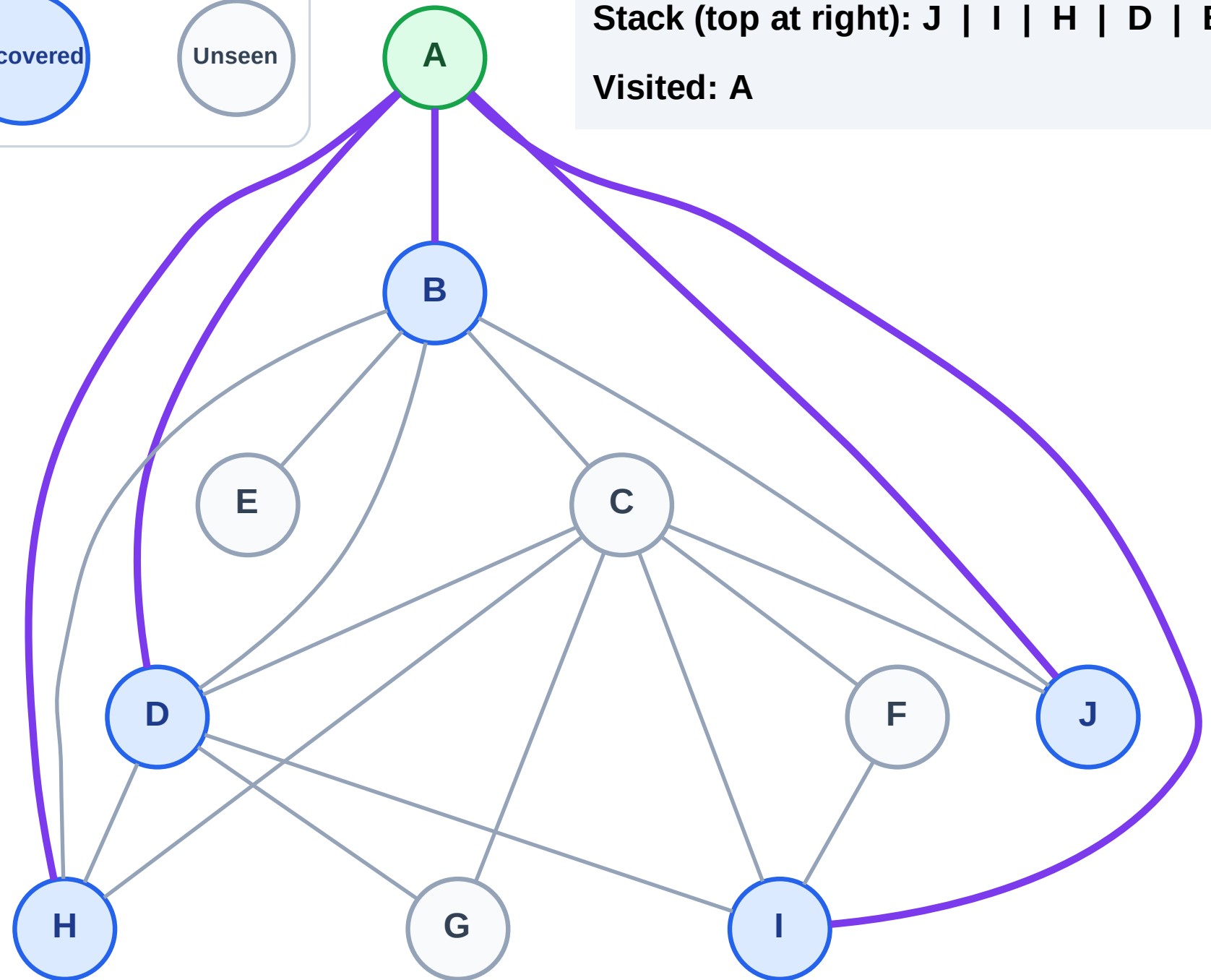
Step 3: Discover J, I, H, D, B from A

Legend



Stack (top at right): J | I | H | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

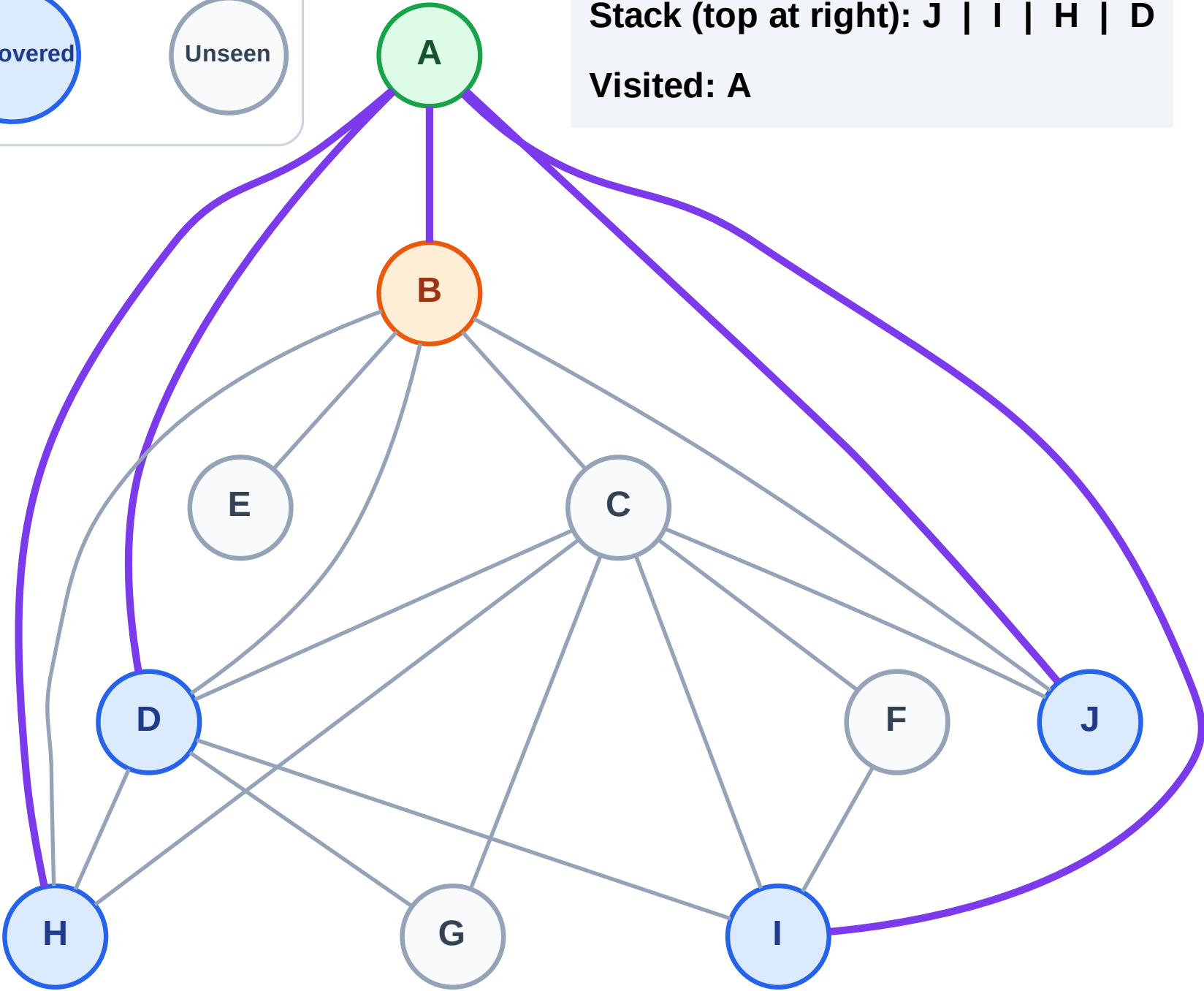
Complete

Processing

Discovered

Unseen

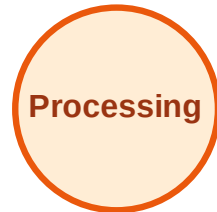
Stack (top at right): J | I | H | D
Visited: A



DFS Traversal

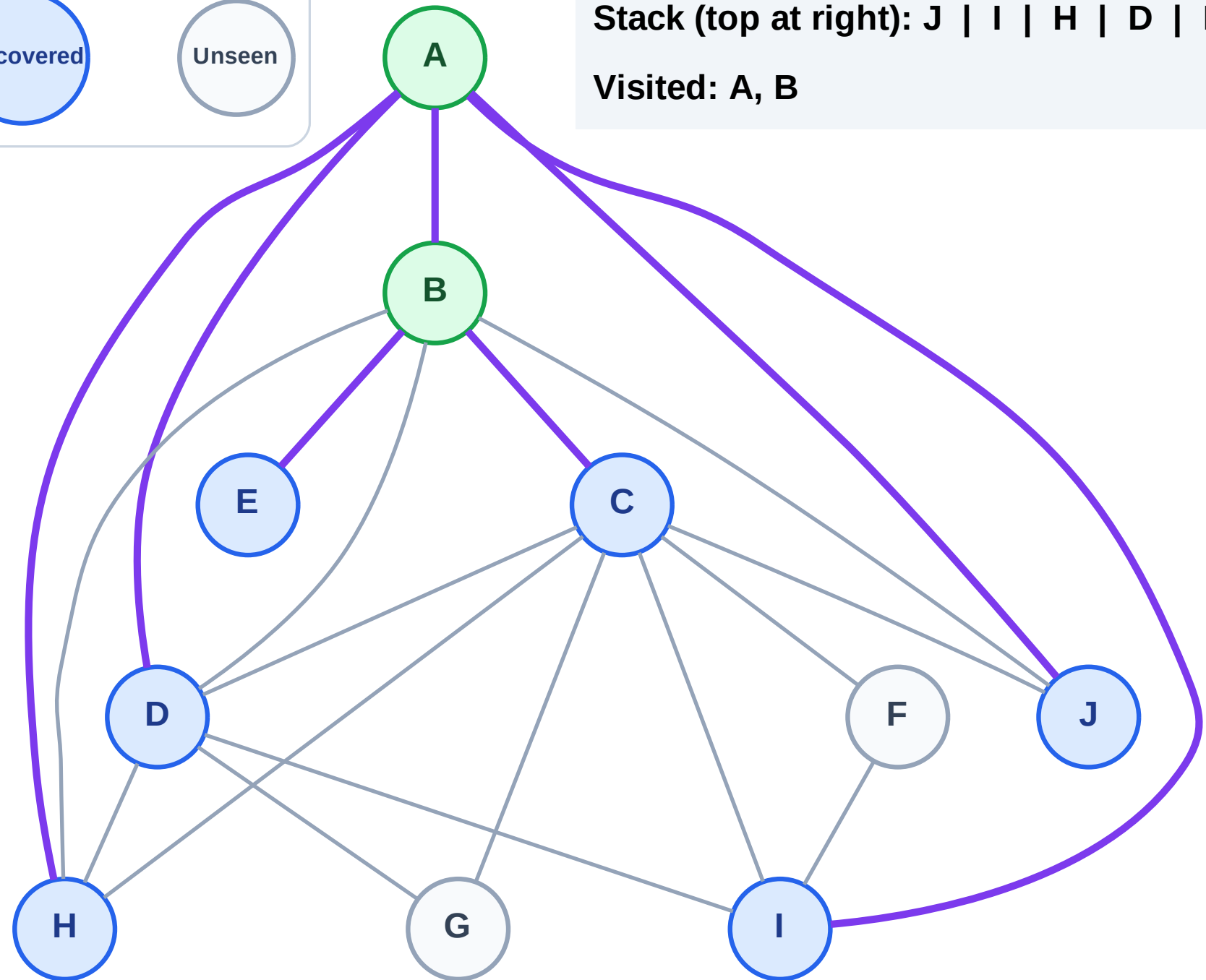
Step 5: Discover E, C from B

Legend



Stack (top at right): J | I | H | D | E | C

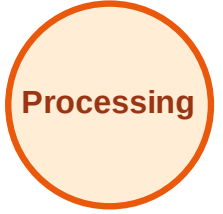
Visited: A, B



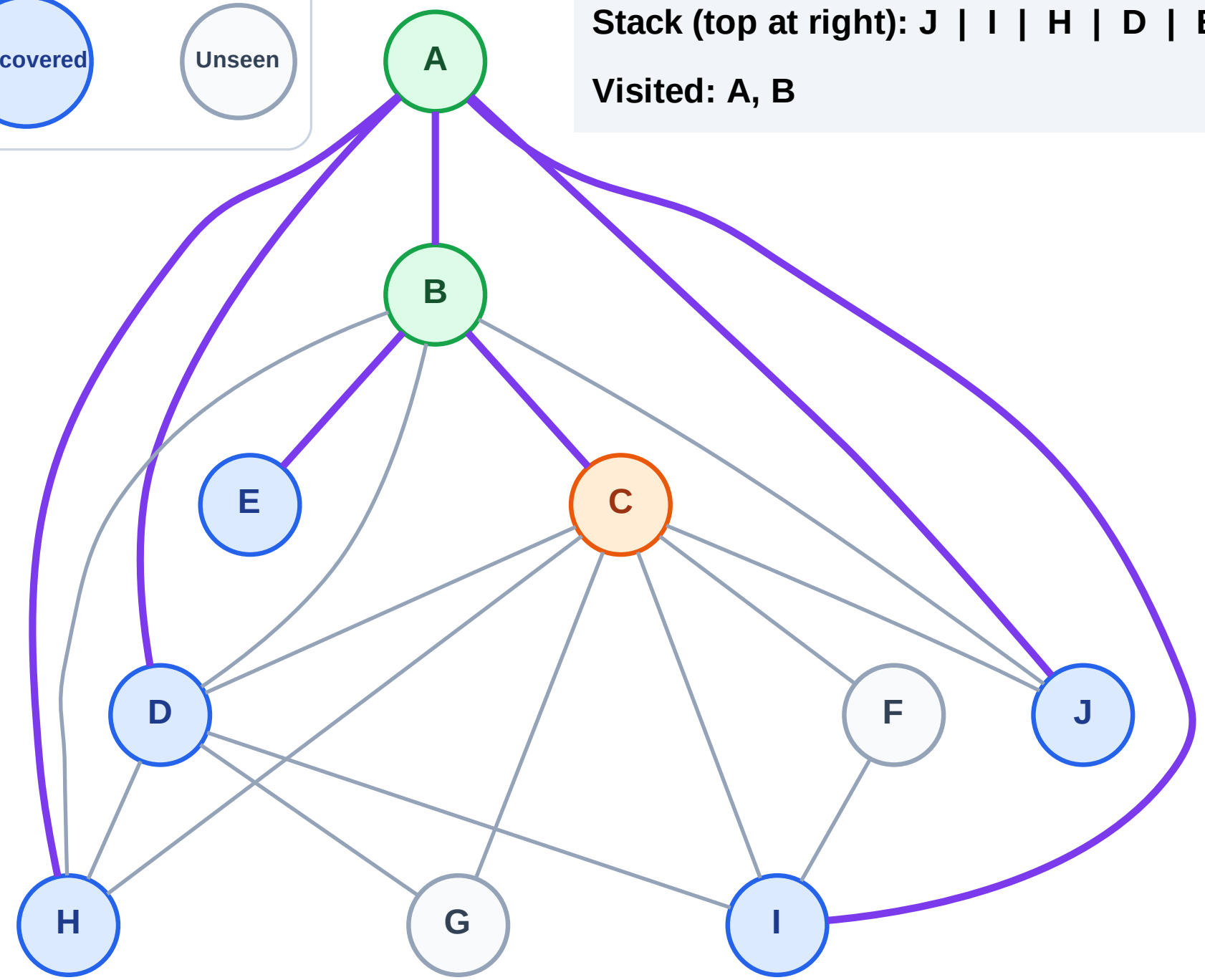
DFS Traversal

Step 6: Process node C

Legend



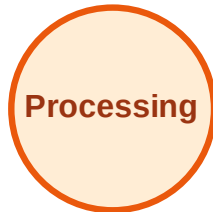
Stack (top at right): J | I | H | D | E
Visited: A, B



DFS Traversal

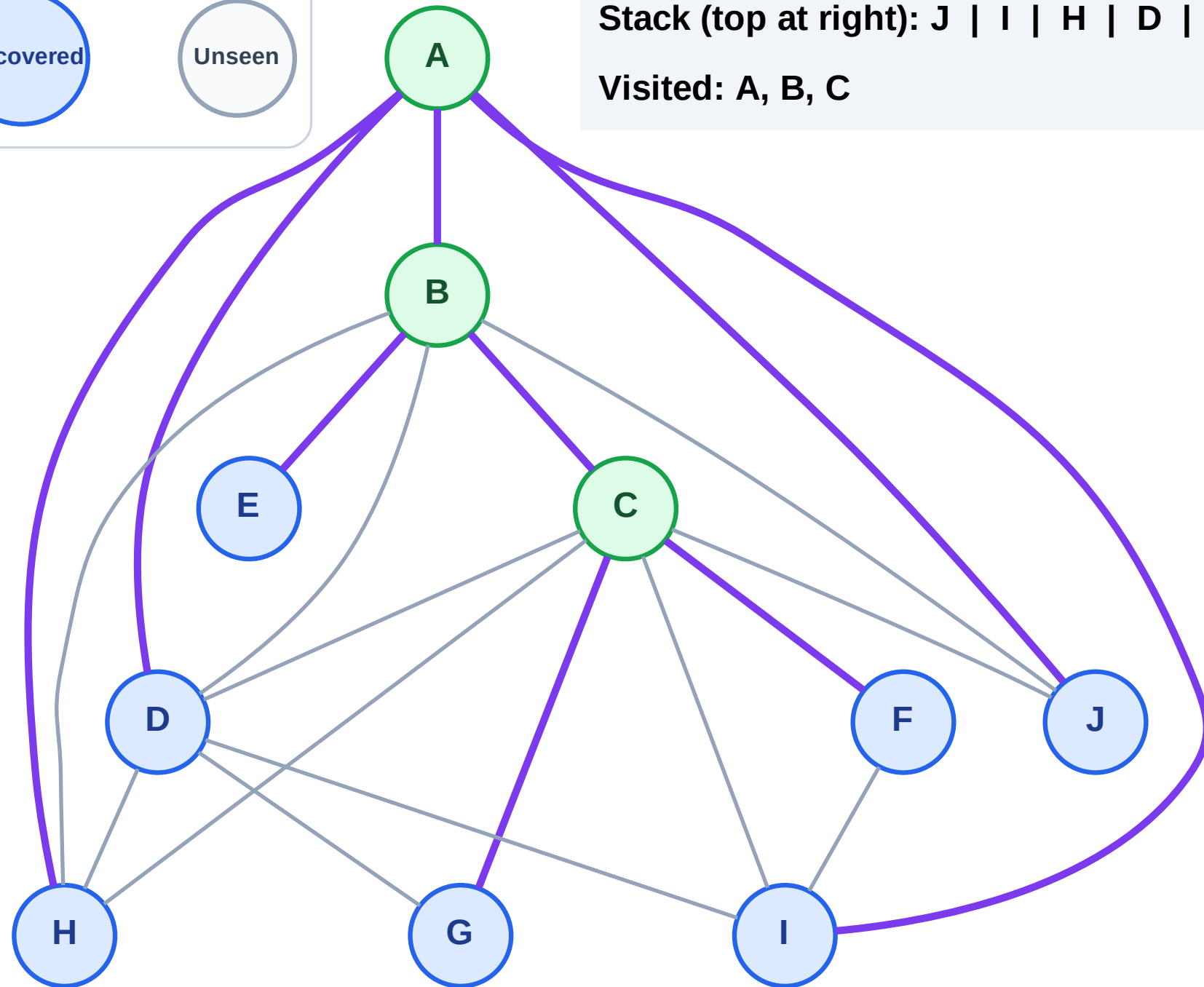
Step 7: Discover G, F from C

Legend



Stack (top at right): J | I | H | D | E | G | F

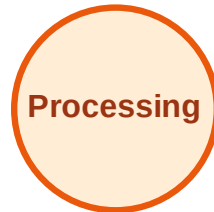
Visited: A, B, C



DFS Traversal

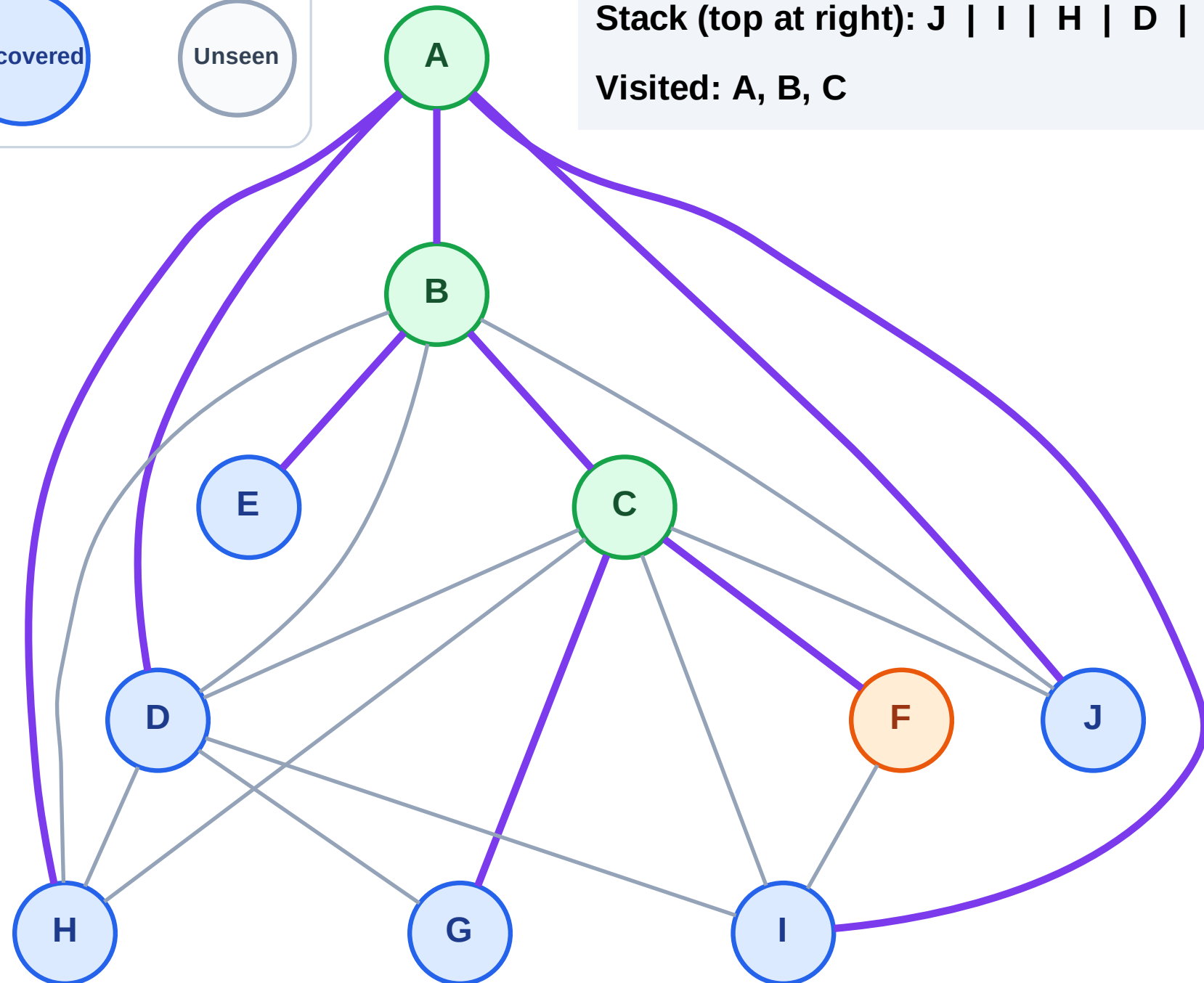
Step 8: Process node F

Legend



Stack (top at right): J | I | H | D | E | G

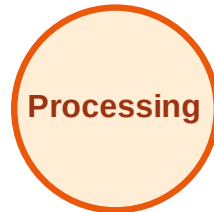
Visited: A, B, C



DFS Traversal

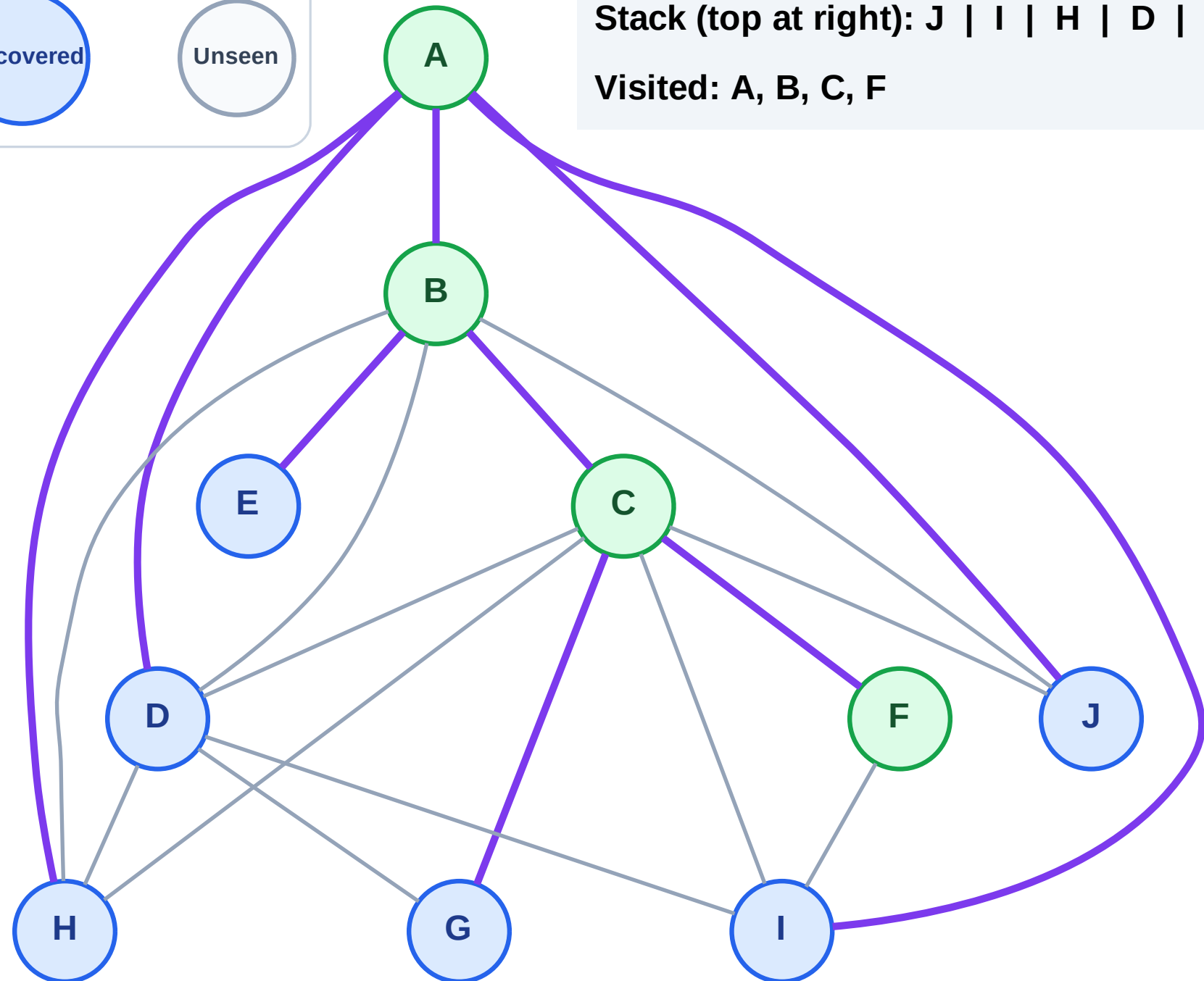
Step 9: No new neighbors from F

Legend



Stack (top at right): J | I | H | D | E | G

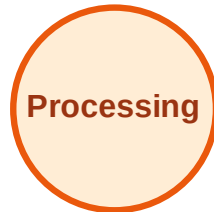
Visited: A, B, C, F



DFS Traversal

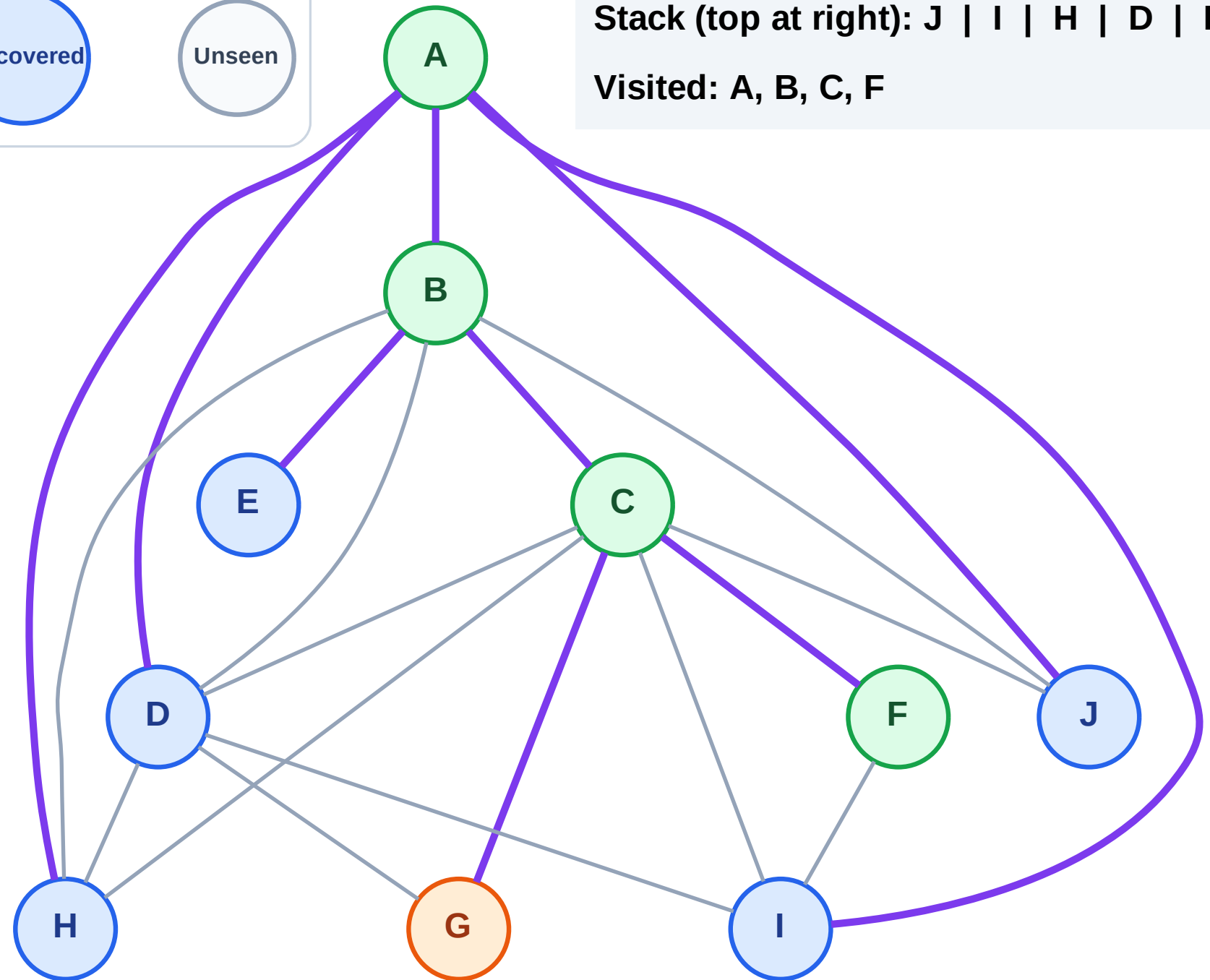
Step 10: Process node G

Legend



Stack (top at right): J | I | H | D | E

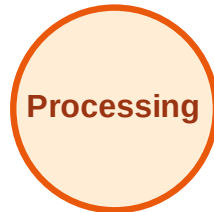
Visited: A, B, C, F



DFS Traversal

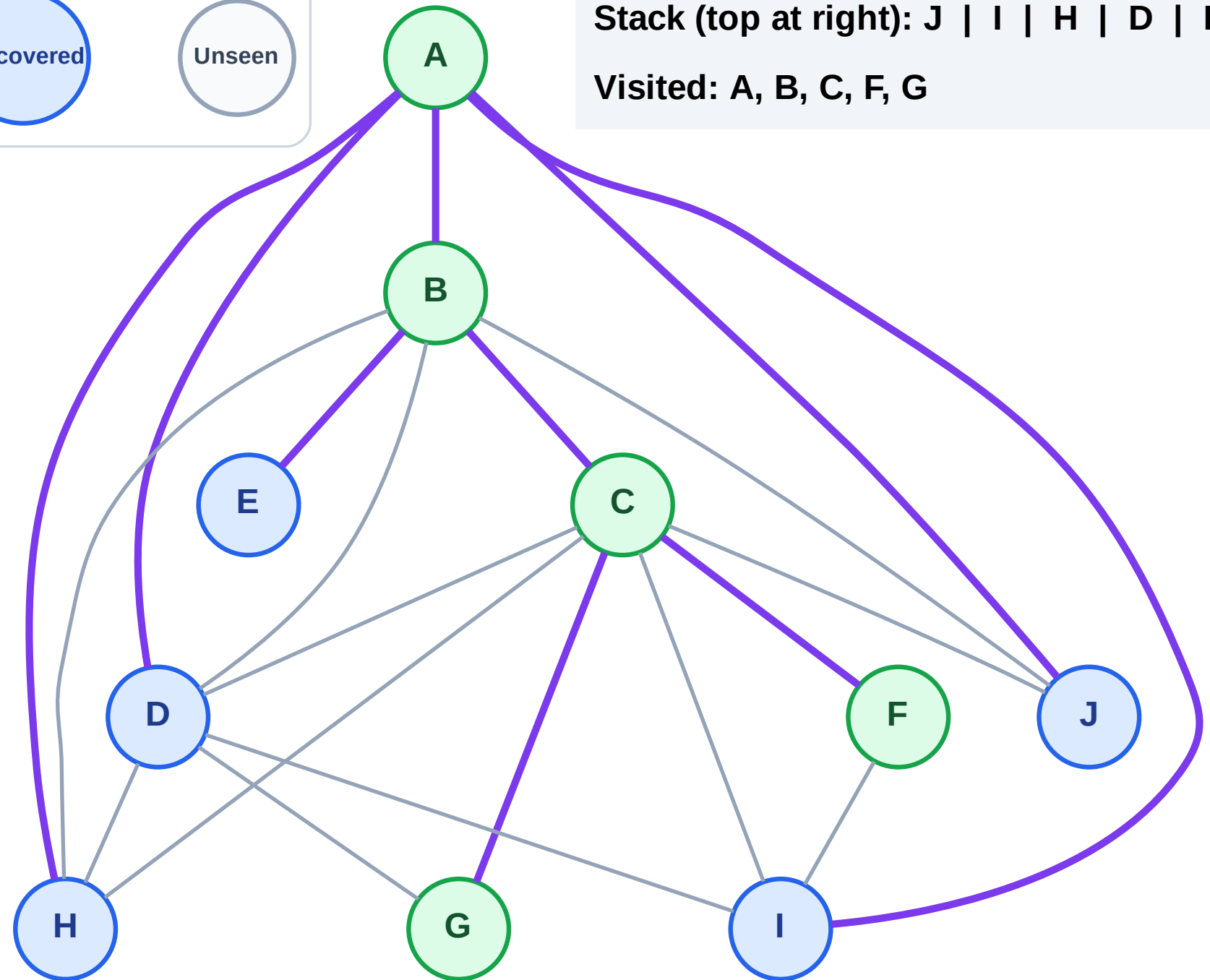
Step 11: No new neighbors from G

Legend



Stack (top at right): J | I | H | D | E

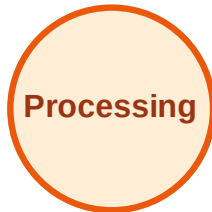
Visited: A, B, C, F, G



DFS Traversal

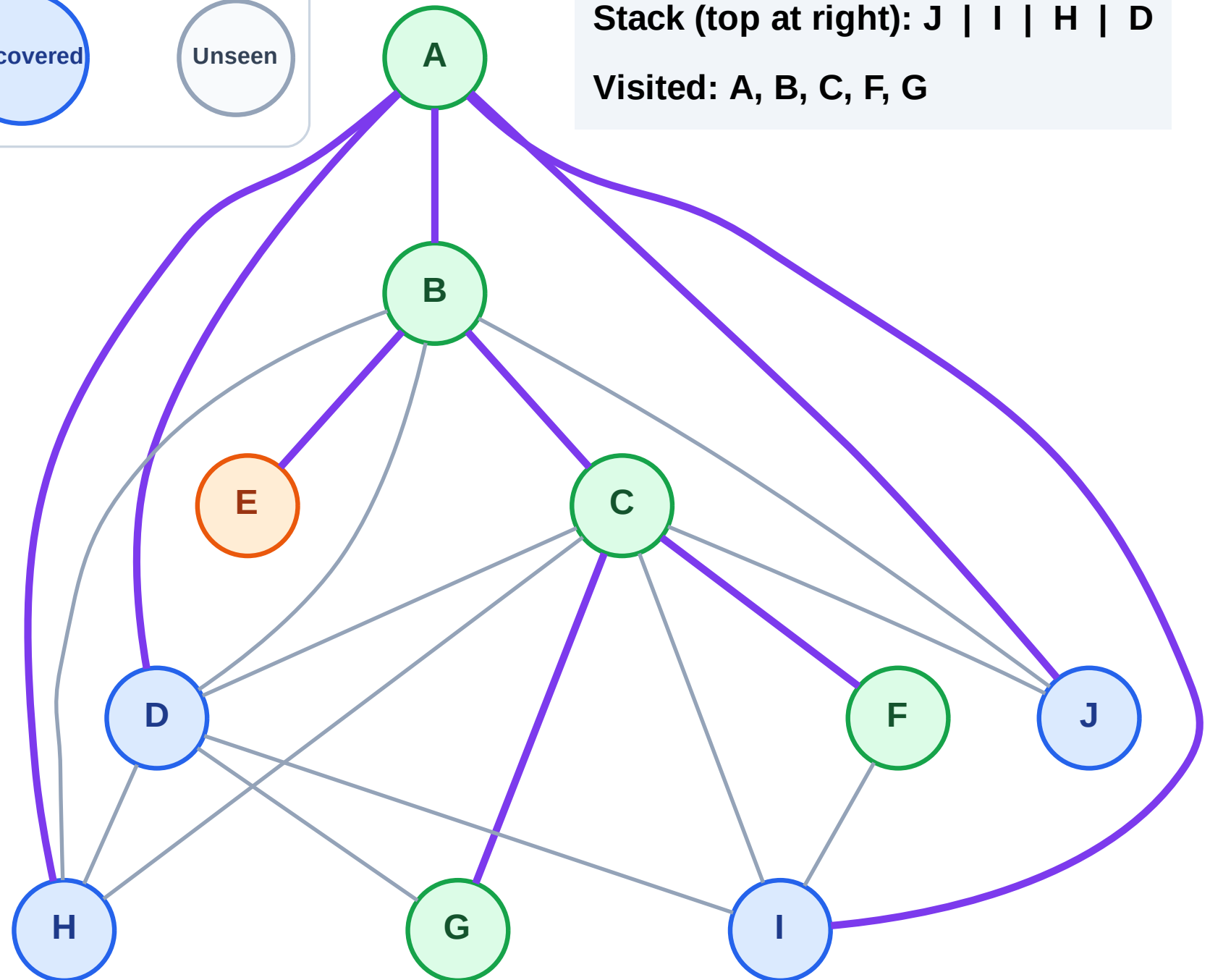
Step 12: Process node E

Legend



Stack (top at right): J | I | H | D

Visited: A, B, C, F, G



DFS Traversal

Step 13: No new neighbors from E

Legend

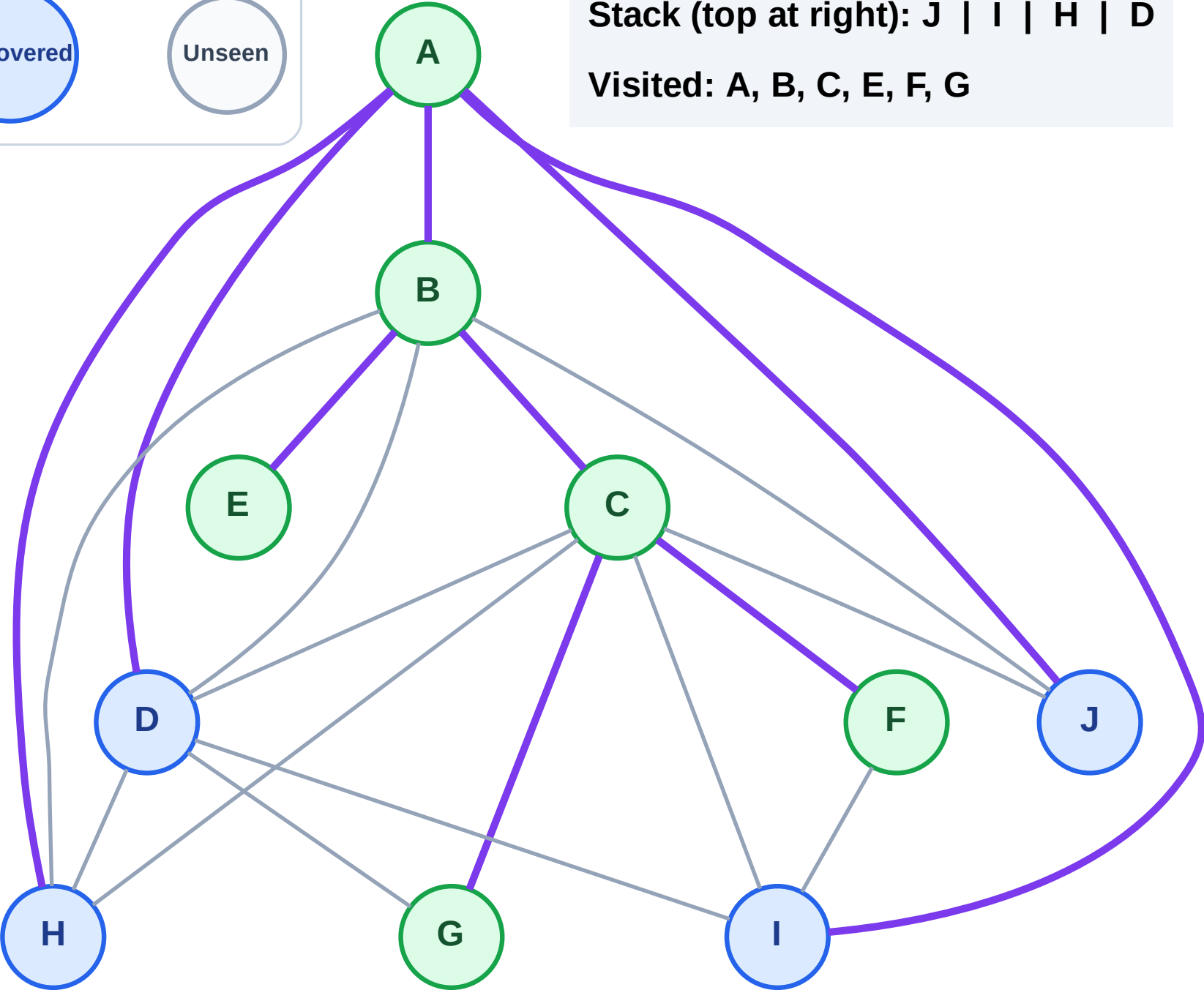
Complete

Processing

Discovered

Unseen

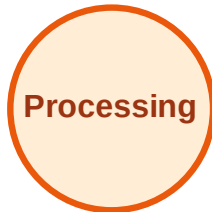
Stack (top at right): J | I | H | D
Visited: A, B, C, E, F, G



DFS Traversal

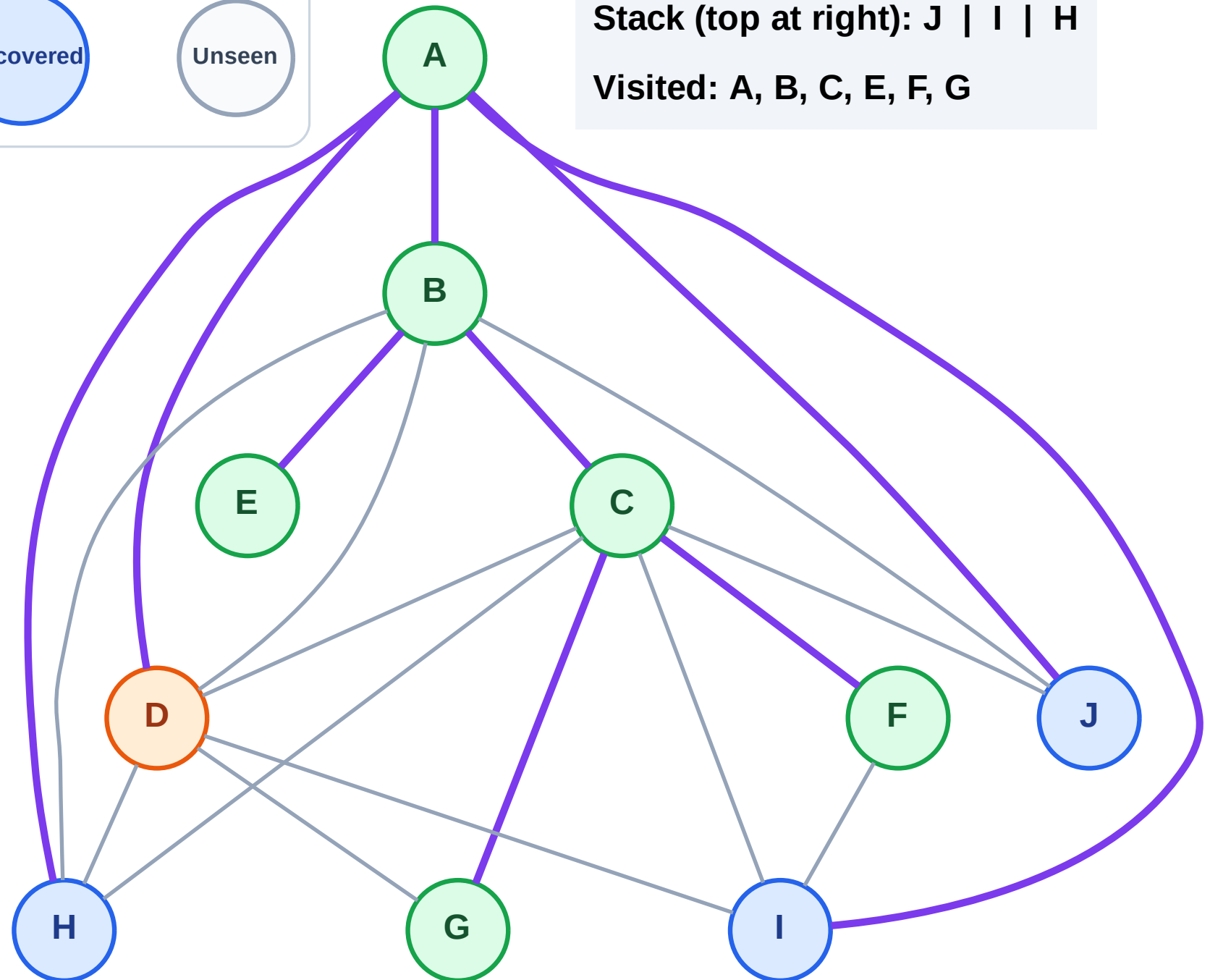
Step 14: Process node D

Legend



Stack (top at right): J | I | H

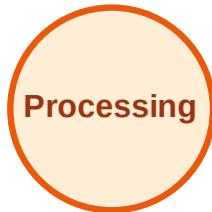
Visited: A, B, C, E, F, G



DFS Traversal

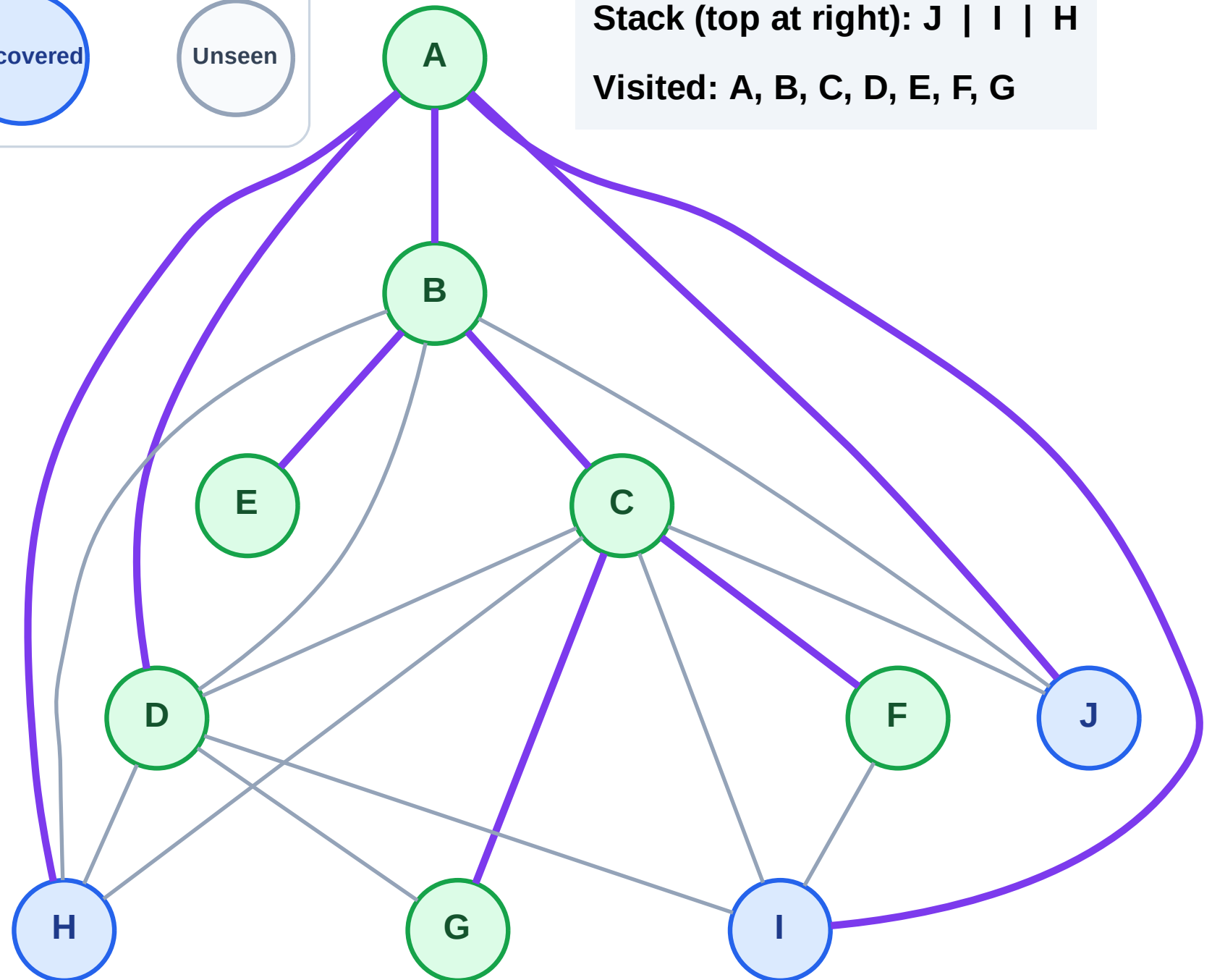
Step 15: No new neighbors from D

Legend



Stack (top at right): J | I | H

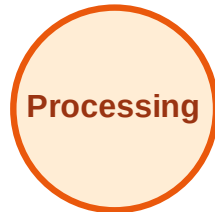
Visited: A, B, C, D, E, F, G



DFS Traversal

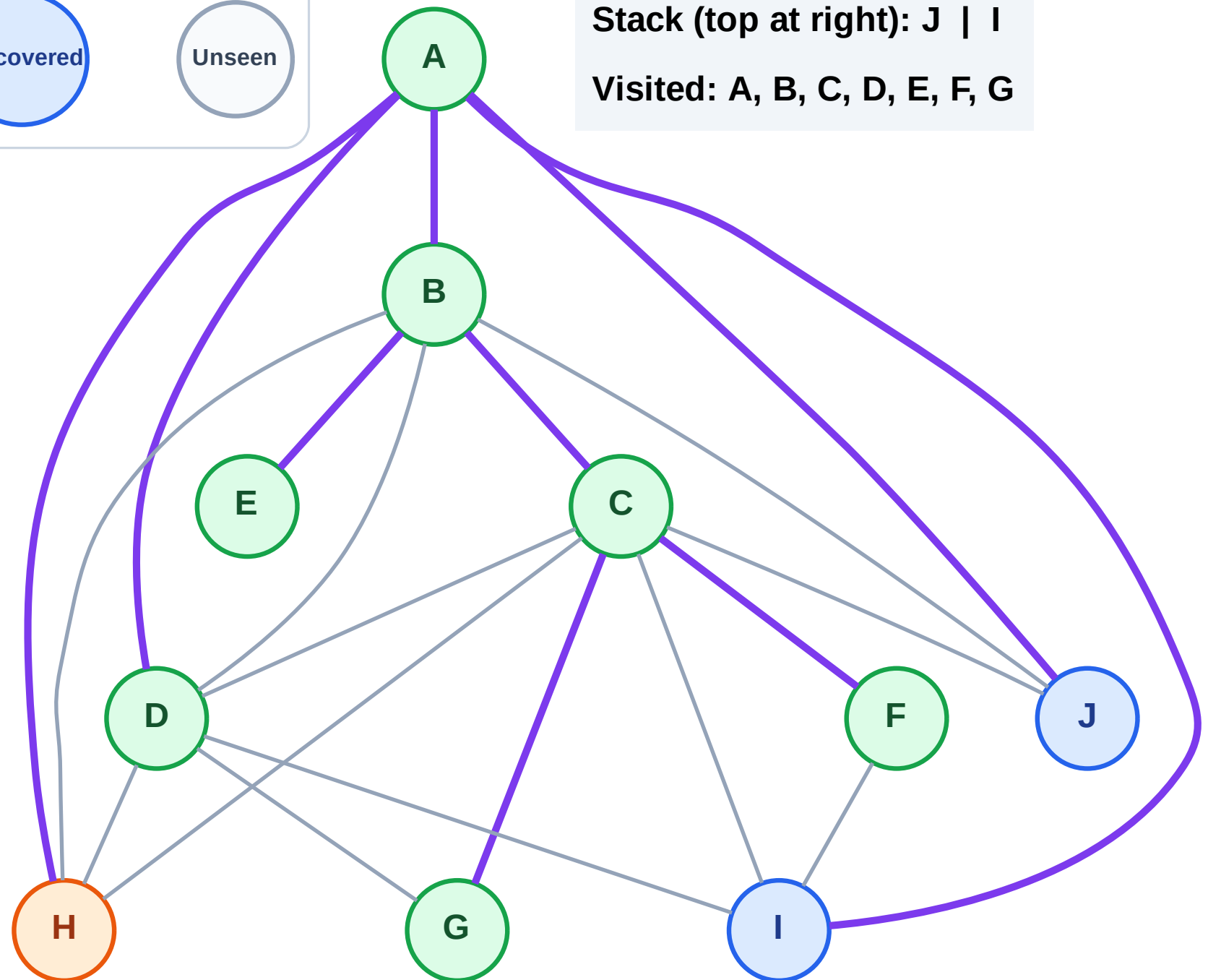
Step 16: Process node H

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G



DFS Traversal

Step 17: No new neighbors from H

Legend

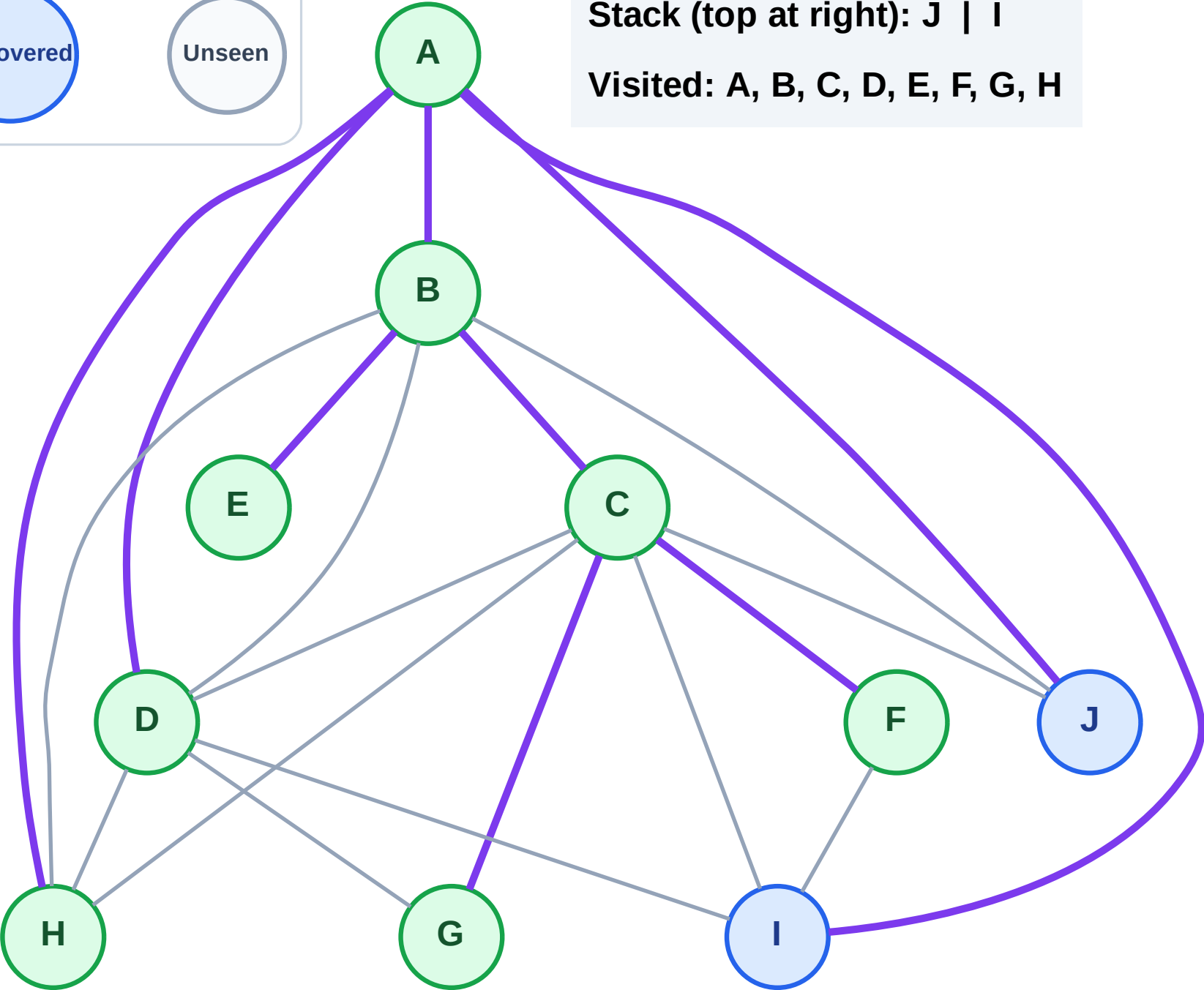
Complete

Processing

Discovered

Unseen

Stack (top at right): J | I
Visited: A, B, C, D, E, F, G, H



DFS Traversal

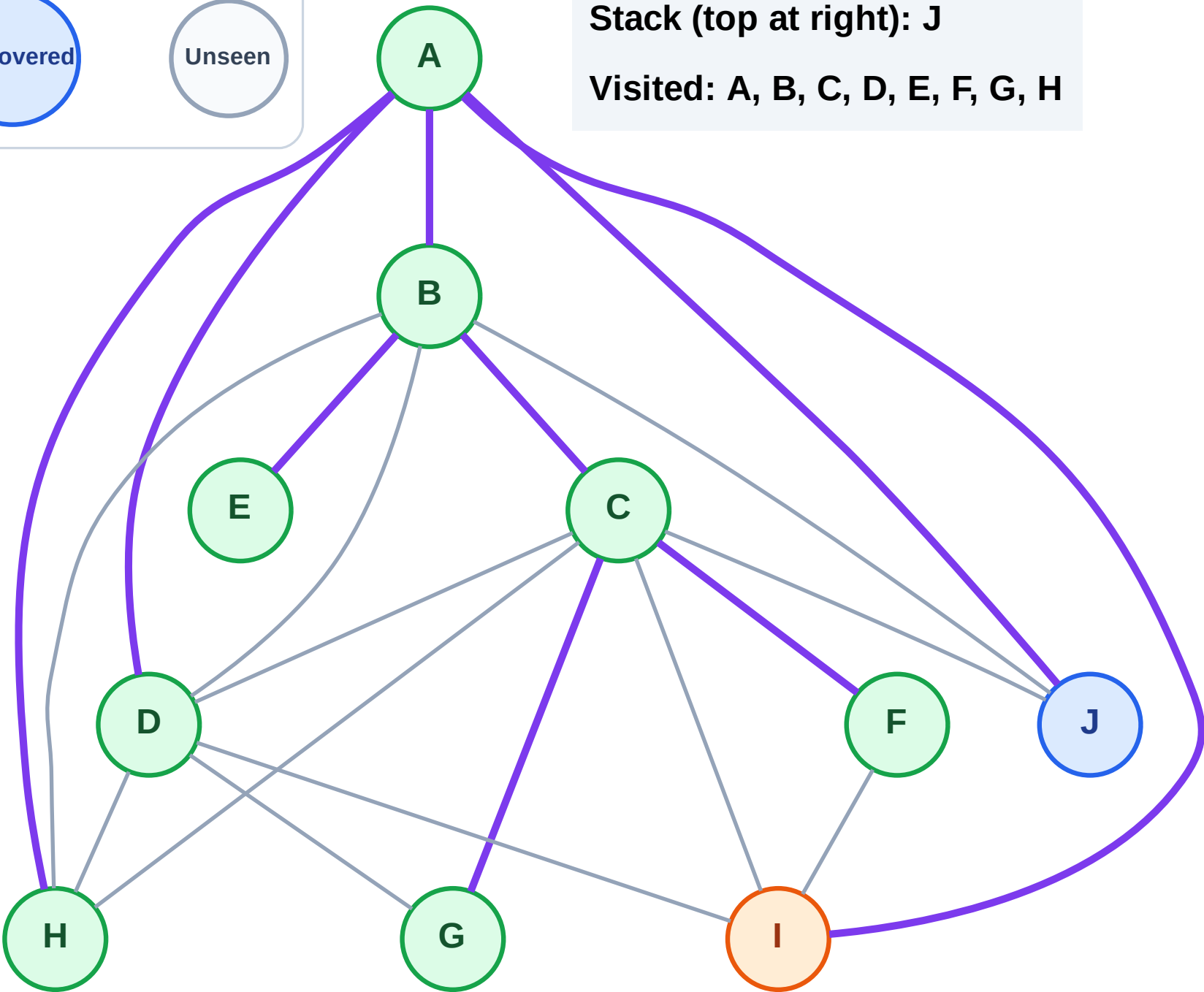
Step 18: Process node I

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



DFS Traversal

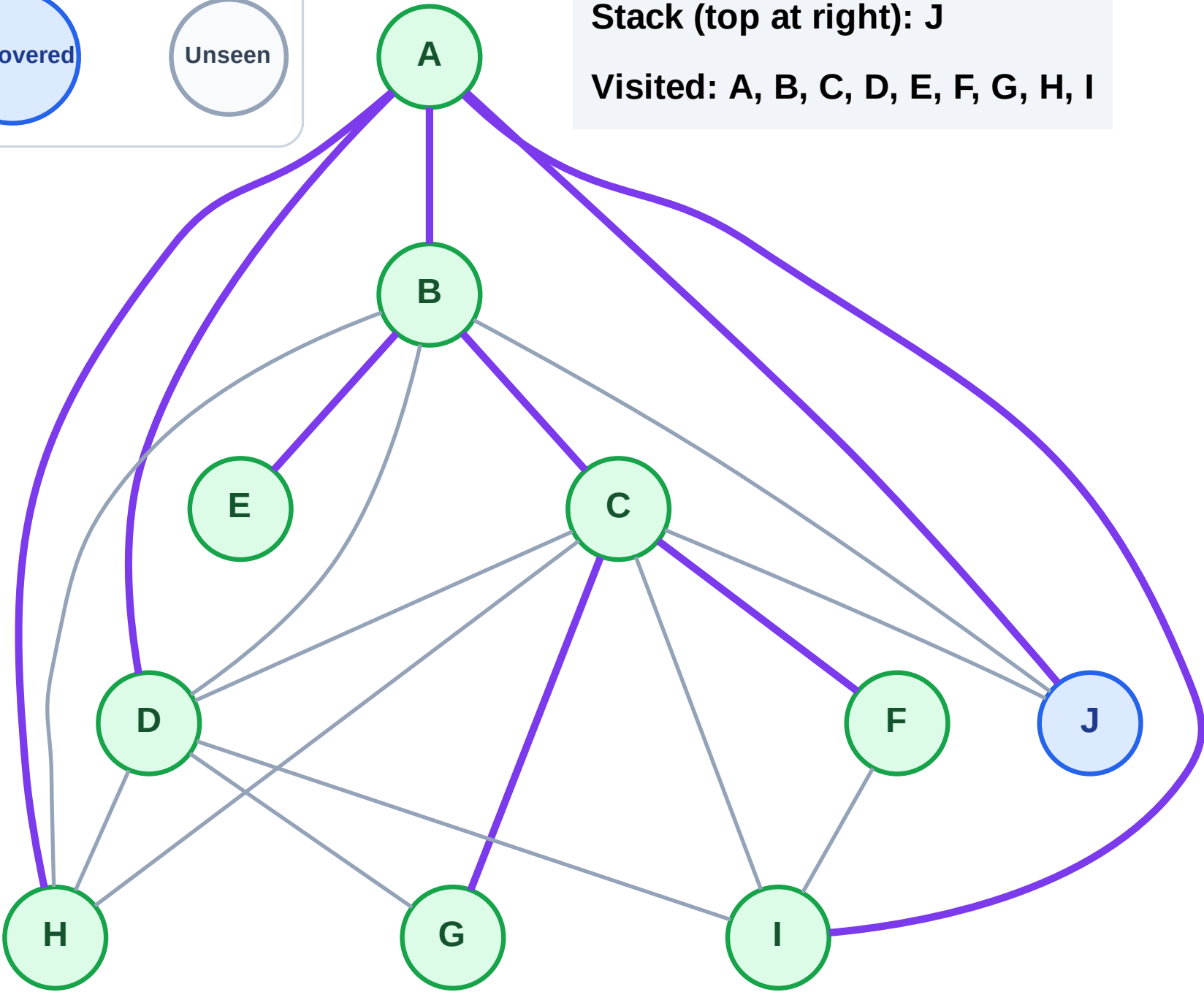
Step 19: No new neighbors from I

Legend



Stack (top at right): J

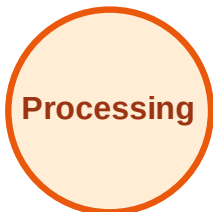
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

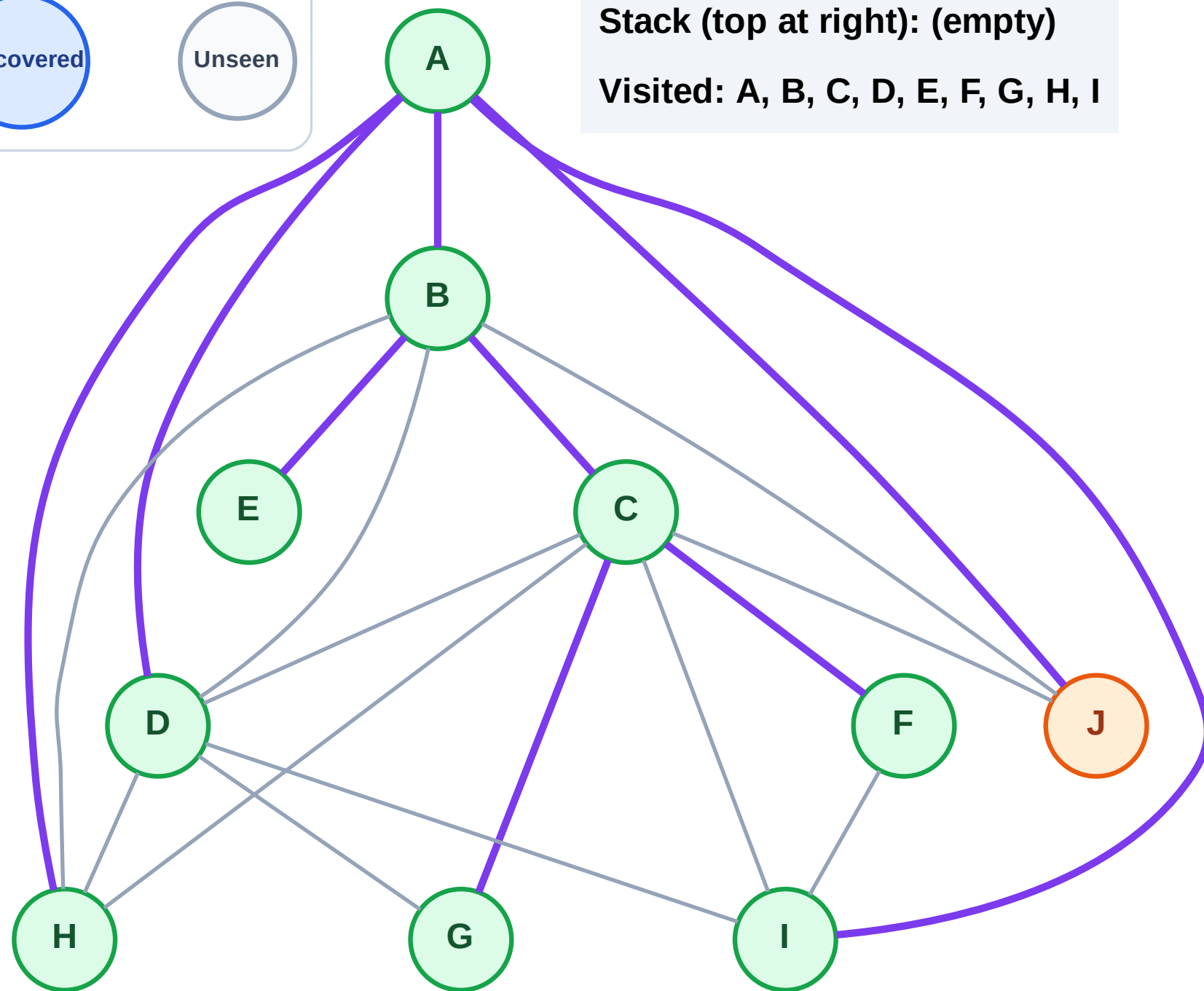
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

